

The SHBT-R Quantum Computer: A Static Holographic Boundary Architecture, its Digital-Twin Simulation Engine, and the Bare-Metal `shbt-os` Microkernel

Abstract

The SHBT-R Quantum Computer is a photonic architecture based on Static Holographic Boundary Theory (SHBT), with WZW affine levels $(k_l, k_q, K) = (26, 8, 312)$ for $SU(2)_{26}$, $SU(3)_8$, and $SO(10)_{312}$. Its processor comprises 312 active InP/InGaAsP microcavity channels arranged in 52 hexamers, with a 72 GHz dynamical-Casimir drive and 36 GHz two-mode squeezed-state seeding. This master specification defines the boundary algebra, discrete holographic reconstruction, optoelectronic dynamics, sapphire/aerogel cryogenic interface, and control thermodynamics. The digital-twin simulation engine couples thermal, photonic, fabrication-yield, and quantum-error-correction models to hardware-in-the-loop control. The bare-metal `shbt-os` microkernel uses a 2112-byte runtime arena aligned to 64 bytes and implements post-quench recovery, SECDED Hamming(72,64), and AVX-512 remapping through a defined memory-mapped interface. SK-9 compilation and fail-closed interlocks complete the control architecture. Requirements V-01–V-35, SECDED Hamming(72,64) ECC, and the SHBT-MMIO-1 register map are primary design and verification standards. The register bank starts at `0x70000000` and spans 56 bytes. The verification and validation matrix defines acceptance criteria, test procedures, and mitigation paths, with software checks and conditional formal verification distinguished from physical qualification. In particular, the encoding shall satisfy $\|W^\dagger W - P_{\text{code}}\|_{\text{op}} \leq 3.430 \times 10^{-3}$, and recovery shall satisfy $t_{\text{rec}} \leq 120.00$ ns under the specified cycle and clock assumptions; device qualification requires calibrated measurements of the corresponding observables and end-to-end timing.

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1 Scope & System Overview

This paper specifies the SHBT-R Quantum Computer, its digital-twin simulation engine, and the bare-metal `shbt-os` control runtime. SHBT denotes *Static Holographic Boundary Theory*; SHBT-R denotes the photonic hardware architecture. The specification covers mathematical assumptions, device parameters, software interfaces, runtime memory, safety behavior, and system acceptance tests.

1.1 Boundary physics and photonic processor

The boundary algebra uses WZW affine levels $(k_l, k_q, K) = (26, 8, 312)$ for $SU(2)_{26}$, $SU(3)_8$, and $SO(10)_{312}$. The framing locks are $K/(2k_l) = 6$ and $K/(3k_q) = 13$, with visible central charge $c_{\text{vis}} = 1325/154$. The processor contains 312 active InP/InGaAsP microcavity channels across 52 six-cavity hexamers, with 24 spare channels in a 336-channel fabrication layout. Affine level and physical channel count are distinct quantities despite the common integer 312. A 72 GHz dynamical-Casimir drive supplies the design’s 36 GHz two-mode squeezed-state seeding stage; coupling to optical microcavity modes requires the frequency-conversion interface defined in Section 3. The cryogenic architecture combines a 4.2 K operating bath, sapphire reservoir, and aerogel acoustic-matching interface.

1.2 Digital twin and deterministic control

The digital twin couples multiphysics thermal and vibration models, photonic coupling and fabrication-yield calculations, and matrix-product-state quantum-error-correction models to a hardware-in-the-loop MMIO interface. The `shbt-os` microkernel supplies post-quench recovery, SECDED Hamming(72,64), and AVX-512 channel remapping. Its zero-heap runtime arena occupies 2112 bytes on a 64-byte boundary; this is the arena size, not the total executable size. SHBT-MMIO-1 defines the register contract, while SK-9 compilation, cycle budgets, and fail-closed interlocks define control execution.

1.3 Implementation repositories

The software implementations are available in two repositories:

- **SHBT Precision Boundary Simulator:**
<https://github.com/sys1own/shbt-precision.git>. Boundary-sector calculations and precision simulation are implemented in Rust/Python [1].
- **SHBT-R Microkernel & Digital-Twin Engine:**
<https://github.com/sys1own/shbt-qc.git>. The engineering simulation, `shbt-os` runtime, HIL tests, and formal-verification tooling share this implementation tree [2].

Appendix A defines code access and build/test entry points. Software paths in this specification are relative to the corresponding repository root.

1.4 Primary design and verification standards

The V-01–V-35 matrix in Section 10 is the primary design and verification standard for all 35 system requirements, including their observables, test protocols, pass/fail criteria, and mitigation paths. Hardware, firmware, and digital-twin implementations shall satisfy each applicable requirement. The following contracts form part of the same normative baseline:

- V-05: the encoding shall satisfy $\|W^\dagger W - P_{\text{code}}\|_{\text{op}} \leq 3.430 \times 10^{-3}$ as defined in Equation 24.
- V-27: complete post-quench recovery shall satisfy $t_{\text{rec}} \leq 120.00$ ns under the complete-stage cycle and clock requirements of Section 9.

- V-34: protected 64-bit payloads shall use SECDED Hamming(72,64) ECC, with decode time $t_{\text{ecc}} \leq 1.20 \text{ ns}$ and correction/detection behavior as specified in Section 8.3.
- SHBT-MMIO-1: the register bank shall use base 0x70000000, fourteen aligned 32-bit words, and a 56-byte span through 0x70000037, with ABI version 1 and the access semantics in Section 8.2.

Appendices B and C supply normative implementation details for this baseline; their supplementary tables shall use the same limits, units, register offsets, and verification criteria as the main specification. Software verification covers functional behavior and formal contracts under explicit model assumptions; physical acceptance additionally requires calibrated metrology, uncertainty bounds, and target-hardware timing. Qualification shall distinguish software and conditional formal results from physical acceptance evidence. After a fallback, the affected requirements and coupled budgets shall be retested against the unchanged acceptance limits.

Specification conventions. “Shall” identifies a mandatory design or verification requirement; nominal values and model predictions describe operating points and shall not replace acceptance limits. A maximum or minimum defines an acceptance limit. Requirements V-01–V-35 in Section 10 apply to the processor, firmware, and digital twin. Frequencies are stated as f or $\omega/(2\pi)$ in Hz; Hamiltonian couplings use angular-frequency symbols. Times use SI units, including the V-27 maximum 120.00 ns and V-34 SECDED Hamming(72,64) ECC maximum 1.20 ns. Register addresses and offsets are hexadecimal in monospaced type; register names and software identifiers also use monospaced type. Bounds shall retain their stated units and numerical precision wherever repeated. All register accesses shall follow SHBT-MMIO-1.

2 Canonical branch, geometry, and physical degrees of freedom

Definition 2.1 (Canonical affine-level vector). *The canonical branch is $(k_l, k_q, K) = (26, 8, 312)$: k_l is the weak-sector WZW affine level of $SU(2)_{26}$, k_q is the color-sector level of $SU(3)_8$, and K is the parent-completion level of $SO(10)_{312}$. These integers are neither spatial/control dimensions nor local Hilbert-space dimensions. The boundary-sector organization is*

$$\mathcal{H}_\partial = \mathcal{H}_{SU(2)_{26}} \otimes \mathcal{H}_{SU(3)_8} \otimes \mathcal{H}_{SO(10)_{312}} \otimes \mathcal{H}_{\text{dark}}.$$

The independent engineering choice is $N_{\text{ch}} = 52 \times 6 = 312$ active channels; its numerical equality to K does not identify affine level with cavity count. The empirical fit is written as $\varepsilon_W^{\text{fit}}(N_{\text{ch}})$ throughout. No physical qubit inventory follows from the levels. Any finite node encoding requires a separate basis, orthogonality measurements, preparation/readout operators, and leakage projector. Six bosonic cavities alone do not imply 18 independently addressable qubits. The fabrication layout includes 336 channels, leaving 24 spares beyond the 312 active channels; activating a spare must preserve the active channel count and update calibration and encoding maps.

2.1 Framing closure, central charges, and discretization

Within the boundary anomaly framing, the scalar obstruction is

$$\Delta_{\text{fr}} = \max \left(\left\| \frac{K}{2k_l} \right\|_{\mathbb{Z}}, \left\| \frac{K}{3k_q} \right\|_{\mathbb{Z}} \right), \quad \|x\|_{\mathbb{Z}} = \min_{n \in \mathbb{Z}} |x - n|, \quad (1)$$

$$I_l^* = 312/52 = 6 \in \mathbb{Z}, \quad I_q^* = 312/24 = 13 \in \mathbb{Z}, \quad (2)$$

$$K = \text{lcm}(52, 24) = 312, \quad \Delta_{\text{fr}} = 0. \quad (3)$$

This proves exact closure of the two stated integer locks, not cancellation of every possible device or field-theoretic anomaly. No relation $26 = 24 + 2$ is an anomaly-cancellation argument. The WZW

formula $c_{G_k} = k \dim G / (k + h_G^\vee)$ gives

$$c_{SU(2)_{26}} = \frac{3(26)}{26+2} = \frac{39}{14}, \quad c_{SU(3)_8} = \frac{8(8)}{8+3} = \frac{64}{11}, \quad (4)$$

$$c_{\text{vis}} = \frac{39}{14} + \frac{64}{11} = \frac{1325}{154} \simeq 8.6039, \quad c_{SO(10)_{312}} = \frac{45(312)}{312+8} = \frac{351}{8} = 43.875, \quad (5)$$

$$c_{\text{dark}}^{\text{comp}} = \frac{834433}{362670} + 1 = \frac{1197103}{362670} \simeq 3.3008. \quad (6)$$

The completed dark value is a boundary-sector ledger input, not the difference between parent and visible central charges, and not a byte-storage capacity. Device qualification shall establish a physical realization of these sectors and their modular completion. Classical actuator coordinates, if used, have a separately calibrated dimension unrelated to k_l . On a two-torus define Voronoi cells and cell-averaged operators by

$$V_k = \{y \in \mathbb{T}^2 : d(y, y_k) \leq d(y, y_j) \ \forall j\}, \quad (7)$$

$$\mathcal{O}_k(\tau) = |V_k|^{-1} \int_{V_k} d^2y \mathcal{O}_{\text{CFT}}(y, \tau), \quad k_c = \pi/a = 2.5113 \times 10^5 \text{ m}^{-1}, \quad a = 12.51 \mu\text{m}. \quad (8)$$

The meaning of the two torus coordinates must be fixed: a Euclidean CFT_2 spacetime torus and a two-dimensional spatial wafer are not interchangeable. The bulk model used below is AdS_3 , with one spatial and one temporal boundary coordinate. A physical two-dimensional wafer may emulate this through a separately specified mapping; it is not itself that spacetime.

The proposed chiral algebra is

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c_{\text{eff}}}{12} m(m^2-1)\delta_{m+n,0}, \quad c_{\text{eff}} := c_{\text{vis}} = 1325/154.$$

The nominal higher-spin residual is $\sum_{s \geq 5} \|W^{(s)}\|^2 = 1.42 \times 10^{-8}$ against V-02's 1.5×10^{-8} target. For the completed nonchiral torus partition function, the modular diagnostic is

$$\delta_{\text{mod}} = \left| \frac{Z(-1/\tau, -1/\bar{\tau}) - Z(\tau, \bar{\tau})}{Z(\tau, \bar{\tau})} \right|.$$

Chiral characters instead transform by a modular matrix. A scalar-weight estimate of $(1.18 \pm 0.03) \times 10^{-7}$ cannot establish this observable without computing Z from the specified sectors. The upper endpoint of that interval reaches 1.21×10^{-7} , above V-03's 1.20×10^{-7} ceiling. No confidence level is given.

3 Optoelectronic dynamics and squeezed-state seeding

3.1 Hexamer Hamiltonian and parameter conventions

Writing all couplings as angular frequencies makes the Hamiltonian dimensionally consistent:

$$H(t) = H_{\text{hex}} + H_{\text{syn}} + H_{\text{cross}} + H_{\text{drive}}(t), \quad (9)$$

$$H_{\text{hex}} = \sum_{m=1}^{52} \sum_{j=1}^6 \hbar \left[\omega_{m,j} a_{m,j}^\dagger a_{m,j} + \frac{\chi}{2} a_{m,j}^{\dagger 2} a_{m,j}^2 - J_{\text{ring}}(a_{m,j}^\dagger a_{m,j+1} + \text{h.c.}) \right], \quad (10)$$

$$H_{\text{syn}} = \sum_{m,j,n} \hbar \left[n \Omega_{\text{EOM}} c_{m,j,n}^\dagger c_{m,j,n} + \kappa_{\text{syn}}(c_{m,j,n+1}^\dagger c_{m,j,n} e^{i\Phi(t)} + \text{h.c.}) \right], \quad (11)$$

$$H_{\text{cross}} = \sum_{\langle m,p \rangle, j} \hbar J_{\text{cross}}(a_{m,j}^\dagger a_{p,j} + \text{h.c.}), \quad (12)$$

$$H_{\text{drive}} = \sum_{m,j} \hbar \left[\Omega_{m,j}(t) e^{-i(\omega_d t + \phi_{m,j}(t))} a_{m,j}^\dagger + \text{h.c.} \right]. \quad (13)$$

The optical resonance is $\omega_0/(2\pi) = 193.41$ THz, approximately 1550 nm; $\chi/(2\pi) = -220.00$ MHz, $J_{\text{cross}}/(2\pi) = 1.15$ MHz, $\kappa_{\text{syn}}/(2\pi) = 12.80$ MHz, and $\Omega_{\text{EOM}}/(2\pi) = 14.50$ GHz. A finite computational truncation is necessary for a negative-Kerr oscillator model. The acceptance baseline specifies $J_{\text{ring}}/(2\pi) = 45.20$ MHz. Device calibration and digital-twin configuration shall use these parameters for V-06–V-08.

For a uniform isolated ring, discrete Fourier modes diagonalize its hopping block:

$$b_{m,q} = 6^{-1/2} \sum_{j=1}^6 e^{-2\pi i q j / 6} a_{m,j}, \quad \omega_q = \omega_0 - 2J_{\text{ring}} \cos(2\pi q / 6).$$

The angular-frequency bandwidth is $4J_{\text{ring}}$, corresponding to 180.80 MHz in ordinary-frequency units. The Kerr propagation estimate is

$$\phi_{\text{Kerr}} = \frac{2\pi}{\lambda} n_2 \frac{P}{A_{\text{eff}}} L_{\text{eff}}, \quad L_{\text{eff}} = \frac{Q\lambda}{2\pi n_g}.$$

The Kerr calibration shall specify n_2 , effective area, group index, intracavity power, and the single-photon normalization used to determine χ . V-08 applies to the calibrated single-photon anharmonicity rather than a classical propagation phase.

3.2 DRAG control and open-system noise

The derivative-removal pulse is

$$\Omega_k(t) = \mathcal{E}_0(t) + i\dot{\mathcal{E}}_0(t)/\chi, \quad (14)$$

$$\mathcal{E}_0(t) = A_0 \left[e^{-(t-t_g/2)^2/(2\sigma_g^2)} - e^{-t_g^2/(8\sigma_g^2)} \right], \quad t_g = 14.21 \text{ ns}, \quad \sigma_g = t_g/4 = 3.5525 \text{ ns}, \quad (15)$$

$$\phi_k(t) = \int_0^t d\tau \mathcal{E}_0(\tau)^2 / (2\chi). \quad (16)$$

The sign follows the chosen drive convention; pulse area, truncation, transfer function, and calibration must also be stated. The nominal leakage is 4.12×10^{-5} per Clifford; verification shall use 10^6 benchmarking cycles with retained measurement data. The proposed memory-kernel dynamics are

$$\dot{\rho}(t) = -\frac{i}{\hbar} [H(t), \rho(t)] + \sum_{i,j} \int_0^t d\tau \mathcal{K}_{ij}(t-\tau) \mathcal{D}[L_{ij}] \rho(\tau), \quad (17)$$

$$\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}, \quad L_{ij} = \sqrt{\gamma_{ij}}(a_i^\dagger a_j + a_j^\dagger a_i), \quad (18)$$

$$\gamma_{ij} = \gamma_0 e^{-|r_i - r_j|/\xi}, \quad J_{\text{bath}}(\omega) = \alpha_{\text{bath}} \omega e^{-\omega/\omega_c}. \quad (19)$$

Model parameters are $\xi = 12.5 \mu\text{m}$, $\Delta\kappa/\kappa = 3.21\%$, $\sigma_T = 4.5 \text{ mK}$, $\alpha_{\text{bath}} = 4.2 \times 10^{-4}$, and $\omega_c/(2\pi) = 4.50 \text{ GHz}$. Positivity and complete positivity of a time-convolution equation are not guaranteed merely by inserting Lindblad-shaped terms: an explicit bath derivation, kernel normalization, initial state, and positivity checks are required. The additional phenomenological spectrum is

$$S_N(f, T, P) = \frac{A_q}{f^{\alpha_q}} + \sum_m \frac{C_m \tau_m}{1 + (2\pi f \tau_m)^2} \frac{\tanh(\hbar\omega_0/2k_B T)}{\sqrt{1 + P_{\text{RF}}/P_{\text{sat}}}} + \frac{S_{\text{carrier}}(T)}{1 + (2\pi f \tau_{\text{carrier}})^2} + S_{\text{mag}}(f). \quad (20)$$

The nominal model inputs are $A_q = 2.1 \times 10^{-11} \text{ e}^2/\text{Hz}$, $\alpha_q = 1.02$, $P_{\text{sat}} = -82.4 \text{ dBm}$, and $\tan \delta_{\text{TLS}} = 1.8 \times 10^{-6}$ in $\text{Al}_x\text{Ga}_{1-x}\text{As}_y\text{Sb}_{1-y}$ cladding. Powers must be converted from dBm to linear units before division; noise terms need a common referred observable and transfer functions before summation.

3.3 Dynamical Casimir effect and conversion requirements

The proposed InP/InGaAs heteroepitaxial substrate modulates an effective boundary at $\Omega/(2\pi) = 72$ GHz, satisfying $\Omega = \omega_a + \omega_b$ with two 36 GHz modes. For a pump phase absorbed into $\xi = re^{i\theta}$,

$$\hat{S}_{ab}(\xi) = e^{\xi^* ab - \xi a^\dagger b^\dagger}, \quad (21)$$

$$|\psi\rangle = \hat{S}_{ab}(\xi)|0,0\rangle = \frac{1}{\cosh r} \sum_{n=0}^{\infty} (-e^{i\theta} \tanh r)^n |n,n\rangle, \quad (22)$$

$$H_{\text{DCE}} = \hbar \chi_{\text{DCE}}(t)(a^\dagger b^\dagger + ab), \quad r = \left| \int \chi_{\text{DCE}}(t) dt \right|. \quad (23)$$

The idealized moving-boundary estimate is $\chi_{\text{DCE}} \simeq \omega_0 x_0/L$, $x(t) = x_0 \sin \Omega t$, and $x_0/L = \Delta n/n$. The designated pump density is $I_p = 895.0 \text{ W cm}^{-2}$. The absorber parameters are $\alpha = 1.2 \times 10^4 \text{ cm}^{-1}$, $\tau_c = 0.85 \text{ ps}$, $\hbar\nu_p = 0.80 \text{ eV}$, and the steady-state estimate $N = \alpha I_p \tau_c / (\hbar\nu_p) \simeq 7.12 \times 10^{13} \text{ cm}^{-3}$. An index-versus-carrier law, optical absorption efficiency, microwave mode volume, pump-window duration, and loaded quality factor are still needed to derive r from I_p . The experimental DCE literature establishes pair generation in superconducting circuits, not this particular semiconductor implementation [8].

Ideal and thermal squeezing. At the target $r = 1.52$, ideal quadrature squeezing is

$$10 \log_{10}(e^{2r}) = 13.2026 \text{ dB}, \quad \sinh^2 r = 4.73827.$$

The thermal occupation at 4.2 K is

$$\bar{n}_{\text{th}} = [e^{h(36 \text{ GHz})/(k_B T)} - 1]^{-1} = 1.96512.$$

Without precooling, the input is not vacuum. In a quadrature convention with vacuum variance $1/2$, a symmetric two-mode squeezed thermal state has $\tilde{\nu}_- = (\bar{n}_{\text{th}} + 1/2)e^{-2r}$; the Gaussian partial-transpose test [11] gives entanglement when

$$r > \frac{1}{2} \ln(2\bar{n}_{\text{th}} + 1) = 0.79769.$$

Its squeezed collective-quadrature variance and suppression relative to vacuum are

$$\frac{V_-}{V_{\text{vac}}} = (2\bar{n}_{\text{th}} + 1)e^{-2r} \simeq 0.2358, \quad -10 \log_{10}[(2\bar{n}_{\text{th}} + 1)e^{-2r}] \simeq 6.27 \text{ dB}.$$

This is not single-mode squeezing of either thermal marginal. Its output occupation is $\bar{n}_{\text{th}} + (2\bar{n}_{\text{th}} + 1) \sinh^2 r$ per mode. The heuristic $\sinh^2 r > \bar{n}_{\text{th}}$ is neither this entanglement criterion nor evidence of a pure two-mode squeezed vacuum. Loss transforms the covariance as $V \mapsto \eta V + (1-\eta)(n_{\text{env}} + 1/2)I$ for equal losses and must be included. At 1550 nm and 4.2 K, $\hbar\nu/k_B T \simeq 2210$; the nominal optical thermal occupation 1.12×10^{-4} is incompatible with equilibrium Bose occupation: $\bar{n}_{\text{th,opt}} \simeq e^{-2210}$ is effectively zero. Optical pump-induced nonequilibrium noise must be measured separately rather than labeled thermal equilibrium.

Microwave-optical interface requirements. The frequency ratio is $193.41 \text{ THz}/36 \text{ GHz} = 5372.5$, approximately 5300; the 72 GHz DCE drive alone cannot bridge it. Adopt two matched, pump-assisted coherent electro-optic converters, one for each squeezed arm, with a 36 GHz microwave resonance, 193.410 THz optical output, and optical pump at $\nu_p = 193.410 - 0.036 = 193.374 \text{ THz}$. An optomechanical alternative must additionally specify the intermediate mechanical frequency, damping, cooling, and thermal-noise transfer. The rotating-wave frequency-conversion model is [3]

$$H_{\text{conv}}/\hbar = G a_o^\dagger b_\mu + G^* a_o b_\mu^\dagger, \quad \nu_o = \nu_p + \nu_\mu.$$

The pump supplies the energy difference; this beam-splitter interaction is not bare microwave absorption by an optical cavity. Adopt the following provisional acceptance contract, not claimed performance of an available transducer:

Quantity	Required specification / test
End-to-end efficiency	$\eta \geq 0.90$ for each arm, including coupling and filters; measure microwave input to optical cavity injection.
Passband	At least 100 MHz full width about 36 GHz and the converted optical carrier; shape and verify the seed spectrum inside the measured passband.
Added noise	$N_{\text{add}} \leq 0.01$ input-referred quadrature-noise quanta, beyond the separately counted vacuum-loss contribution; measure with the conversion pump on.
Phase and mode matching	Common optical reference for both pumps and homodyne LO; rms squeezed-quadrature angle error $\sigma_\theta \leq 0.01$ rad; calibrate differential delay.
Thermal staging	Generate the optical pump outside 4.2 K; measure cold absorption, RF loss, and converter electronics together against the remaining 30.56 mW V-25 margin.
End-to-end qualification	Reconstruct the two-arm covariance including phase noise, loss, pump leakage, and mode mismatch; demonstrate sub-vacuum EPR variance and entanglement.

For matched arms define N_{add} by $V_{\text{out}} = \eta V_{\text{in}} + (1 - \eta)I/2 + \eta N_{\text{add}}I$. For small independent angle jitter the normalized squeezed variance is approximately

$$v_{\text{out},-} = \eta[v_- + (v_+ - v_-)\sigma_\theta^2] + (1 - \eta) + 2\eta N_{\text{add}}, \quad v_{\pm} = (2\bar{n}_{\text{th}} + 1)e^{\pm 2r}.$$

At the stated limiting targets this is about 0.34 (about 4.7 dB suppression), illustrating that conversion does not restore the ideal 13.20 dB. Unmatched loss and correlated noise require the full covariance, not this scalar estimate. Device geometry, pump power, coupling rates, and measured transfer functions remain release gates. The stated time-averaged incident pump power is

$$52(1.60 \times 10^{-5} \text{ cm}^2)(895.0 \text{ W cm}^{-2})(0.10) = 74.464 \text{ mW}.$$

Only its absorbed, cold-stage fraction belongs in the heat budget; it cannot be counted twice.

4 Cryogenics, fabrication, and supporting hardware

4.1 Sapphire reservoir and acoustic matching

The prescribed reservoir is single-crystal sapphire at 4.2 K, with $\rho = 3980 \text{ kg m}^{-3}$, longitudinal speed $c_L = 11100 \text{ m s}^{-1}$, and $Z = \rho c_L = 44.178 \text{ MRayl}$. For $\Theta_D = 1047 \text{ K}$ and atomic density $n = 5N_A\rho/M$, $M = 101.96 \text{ g mol}^{-1}$, the low-temperature Debye estimate is

$$c_v(T) = \frac{12\pi^4}{5}nk_B(T/\Theta_D)^3 \simeq 2.448 \times 10^{-5} \text{ J cm}^{-3}\text{K}^{-1} \quad (T = 4.2 \text{ K}).$$

For a specified hold energy $Q = P\tau$ and small ΔT ,

$$V \geq \frac{P\tau}{c_v\Delta T}.$$

The design inputs $P = 84.20 \text{ mW}$, $\tau = 1.000 \text{ ms}$, and $\Delta T = 1.800 \text{ mK}$ yield approximately 1911 cm^3 . The adopted *minimum design volume* remains $V \geq 1911.0 \text{ cm}^3$; it is not derived from the corrected line-switching power in Section 6. The proposed $124 \times 124 \times 124.3 \text{ mm}^3$ boule is about 1.911 liters and 7.61 kg. A heat-capacity calculation does not establish nanosecond thermal evacuation: the acoustic crossing time of 124 mm at c_L is about $11.2 \mu\text{s}$. The volume dynamically accessible during a quench depends on contact geometry, diffusion, and phonon transport.

For the helium approximation $\rho_{\text{He}} = 125 \text{ kg m}^{-3}$ and $c_{\text{He}} = 240 \text{ m s}^{-1}$, $Z_{\text{He}} = 0.030 \text{ MRayl}$. Normal-incidence plane-wave power transmission at an ideal interface is

$$\mathcal{T} = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2} \simeq 2.71 \times 10^{-3}.$$

For a lossless layer of thickness d_m and wave number q_m , its input impedance is

$$Z_{\text{in}} = Z_m \frac{Z_2 + iZ_m \tan(q_m d_m)}{Z_m + iZ_2 \tan(q_m d_m)}.$$

At $q_m d_m = \pi/2$, $Z_{\text{in}} = Z_m^2/Z_2$; matching $Z_{\text{in}} = Z_1$ gives

$$Z_m = \sqrt{Z_{\text{sap}} Z_{\text{He}}} = 1.15123 \text{ MRayl}, \quad d_m = \frac{c_m}{4f_m} = 6.395 \text{ nm} \quad (\text{design target}).$$

The proposed layer is nanoporous silica aerogel. Its density, sound speed, dispersion, and matching frequency must jointly satisfy these equations; thickness and impedance alone do not specify it. The matching-layer design uses $f_m = 72 \text{ GHz}$, $c_m = 1841.8 \text{ m s}^{-1}$, and $\rho_m = 625 \text{ kg m}^{-3}$, which are consistent with the target thickness and impedance to rounding. Picosecond-ultrasonic characterization shall establish the acoustic parameters. Matching at 36 GHz with the same thickness would require a different material. An 850 nm optical aerogel coating and a 6.394 nm mean pore diameter are distinct quantities, not alternative measurements of this 6.395 nm acoustic layer.

The broadband thermal conductance requires integration over phonon frequency, polarization, and angle. A monochromatic quarter-wave optimum cannot establish broadband heat removal. Near equilibrium one may write $q = G_K \Delta T$, with $G_K \simeq c_v \bar{c} \langle \mathcal{T} \rangle / 4$ in a simplified mismatch model. An alternate interface model uses $G_K(T) = 142.0 T^3 \text{ W m}^{-2} \text{ K}^{-1}$; its integrated flux law is $q = (142.0/4)(T_1^4 - T_2^4) \text{ W m}^{-2}$ when temperatures are in kelvin.

4.2 Five-step process and statistical process control

Table 1: Fabrication recipe and design parameters; process qualification pending.

Step	Process	Tool/chemistry	Critical settings
1	MOCVD epitaxy	Aixtron Crius; TMIn, TEGa, AsH ₃ , PH ₃ , CBr ₄ (p), SiH ₄ (n)	640.0 °C, 50 mbar, 1.05 nm/s $\pm$ 0.5%
2	Electron-beam lithography	Vistec EBPG5200, 100 kV, HSQ Fox-16	380 μ C/cm ² dose, 500 pA beam
3	Low-damage ICP-RIE	Oxford PlasmaPro 100; Cl ₂ /CH ₄ /Ar	12/8/4 sccm; RF 45 W, ICP 650 W, 180 °C
4	Sol-gel coating	Methyltrimethoxysilane; spin coater	3500 rpm, 45 s, optical coating 850 $\pm$ 5 nm
5	Supercritical drying	Tousimis Automegasamdri-916B; liquid CO ₂	38.5 °C, 8.5 MPa, release 0.15 MPa/min

The wafer specification is a 100 mm semi-insulating InP(100) substrate.

SPC metric	Target	Nominal mean	Nominal 3 σ	C_{pk}
Aerogel index	1.0180	1.0182	$\pm$ 0.0008	1.74
Pore diameter (nm)	6.400	6.394	$\pm$ 0.005	1.81
Waveguide CD (nm)	250.00	250.12	$\pm$ 0.42	1.69
Sidewall angle (degrees)	90.00	89.88	$\pm$ 0.04	1.92
PL peak (nm)	1550.00	1549.80	$\pm$ 0.58	1.65

The process capability index $C_{pk} = \min(\text{USL} - \mu, \mu - \text{LSL})/(3\sigma)$ requires separate limits and sample statistics; the table alone does not establish these capability values. The nominal cooldown stress results are $\tau_{\max} = 1.12$ MPa and $G_I = 2.15$ J/m² against $G_{Ic} = 3.65 \pm 0.12$ J/m², giving a central-value safety ratio 1.70.

4.3 Magnetic enclosure, interposer, and vibration

The four-tier enclosure is two 1.5 mm Cryoperm-10 shells ($\mu_r = 120000, 145000$; nominal attenuations 42.1, 48.3 dB), a 3.0 mm five-nines aluminum eddy-current shell (22.4 dB at 40 GHz), and a 2.0 mm YBCO cylinder. Its proposed critical-state law is

$$J_c(B) = \frac{J_{c0}}{1 + |B|/B_0}, \quad J_{c0} = 2.4 \times 10^6 \text{ A/cm}^2, \quad B_0 = 0.85 \text{ T}.$$

The enclosure specifies 364 waveguide penetrations, $L/D \geq 5.2$, and a 145 GHz cutoff. Its nominal field budget reduces a 45.01 μ T external field to 0.0812 nT, versus V-21's 0.0975 nT maximum. That field ratio corresponds to approximately 114.9 dB. Qualification shall measure DC field suppression and RF penetration attenuation separately, with frequency-dependent transfer functions for each shell and penetration. V-21 acceptance shall use the measured residual field in the active volume.

12-layer RO4350B/RO4450F interposer	Simulated	Nominal	Limit
Z_0 (Ω)	50.08	50.12	50.12 ± 0.80
S_{21} (dB/cm)	-0.11	-0.12	≥ -0.18
S_{11} (dB)	-28.4	-27.1	≤ -22.0
FEXT (dB)	-74.2	-72.4	≤ -70.0
Power-loop area (mm ²)	0.010	0.012	≤ 0.025
Clock skew (ps)	0.95	1.18	≤ 2.50
PDN impedance (m Ω)	14.2	16.8	≤ 25.0 (DC-10 GHz)

The impedance and FEXT limits shall satisfy V-23 and V-24 respectively. Vibration dynamics use

$$M\ddot{x} + C\dot{x} + Kx = F_{\text{PT}} + F_{\text{harness}} + F_{\text{piezo}}, \quad H(\omega) = (-\omega^2 M + i\omega C + K)^{-1}, \quad x_{\text{rms}}^2 = \int_1^{500} S_x(f) df.$$

The baseline vibration model uses harness stiffness 1.42×10^3 N/m, 35.5 dB active rejection, and $x_{\text{rms}} = 0.62$ nm, below V-28's 0.85 nm target. The phase transfer $2\pi n_{\text{eff}} x_{\text{rms}} / \lambda$, with $n_{\text{eff}} = 3.24$, gives 8.14×10^{-3} rad. Qualification shall account for this phase excursion relative to the $2\pi/2^{16} = 9.587 \times 10^{-5}$ rad DAC step. An alternative rejection model uses 37.4 dB, reducing 4.04 rad to 54.3 mrad; its calibration shall be specified independently. Each configuration shall provide full transfer functions, force/displacement spectral units, and piezo stability margins.

4.4 Cold-stage heat balance and electronics

Stage heat source	Nominal mW	Nominal percent
312 optical resonators	15.34	1.87
28 nm FD-SOI driver array	412.00	50.28
Cryogenic MPS decoder	285.00	34.78
364 optical-fiber harness lines	18.20	2.22
RF coaxial harness	34.60	4.22
Phosphor-bronze DC leads	12.80	1.56
Multilayer-insulation radiation	41.50	5.06
Total	819.44	100.00

The specified 1.500 W cooling capacity gives a 680.56 mW (45.37%) reserve for that table; the nominal 1.380 W runaway threshold gives a 560.56 mW margin. Adding the separately specified 74.464 mW DCE pump as an additional fully absorbed load would give 893.904 mW: V-25 then fails its 850 mW ceiling even though the 40% capacity-reserve criterion barely passes. The acceptance baseline includes the full 74.464 mW pump allocation *inside* the 412.00 mW driver row, conservatively treating the incident duty-averaged pump as fully absorbed at 4.2 K. The remaining driver/electronics allocation is $412.00 - 74.464 = 337.536$ mW. Thus the baseline remains 819.44 mW, with $850.00 - 819.44 = 30.56$ mW unallocated for the new conversion interface and any other omitted cold load. If a separate 412.00 mW electronics load cannot be reduced or restaged, the additive 893.904 mW branch fails V-25 by 43.904 mW and is not accepted.

Front-end block	Key performance	mW/channel	Count
16-bit, 1.2 GSa/s DAC	SFDR 84.5 dBc	1.12	312
40 GSa/s pulse modulator	Rise time 8.2 ps	2.45	52
12-bit, 2.5 GSa/s ADC	ENOB 10.82 bits	3.10	104
SiGe transimpedance amplifier	Gain 82 dB Ω	0.85	312

These four arrays total 1064.44 mW if continuously active at 4.2 K. The specified *duty-cycled acceptance mode* assigns all four arrays to 4.2 K, caps each active duty factor at 0.25 over every validated thermal averaging window, and allocates as follows:

Block (4.2 K)	Continuous mW	Duty factor	Active-average mW
DAC array	349.44	0.25	87.36
Pulse modulator array	127.40	0.25	31.85
ADC array	322.40	0.25	80.60
TIA array	265.20	0.25	66.30
Front-end subtotal	1064.44		266.11
Idle, wake-up, bias and other driver overhead			≤ 71.426
DCE absorbed pump (incident duty 0.10)			74.464
Driver allocation			≤ 412.000

For each block the actual contribution is $d_i P_{i,\text{on}} + (1 - d_i) P_{i,\text{idle}} + E_{i,\text{wake}} f_{i,\text{wake}}$; idle and wake-up terms are charged to the 71.426 mW overhead, not assumed zero. The corrected 0.08408 mW line-switching estimate is a suballocation of this driver budget, not an extra load or an independent measurement of the front-end subtotal. The DCE optical pump duty and electronics duty are distinct. Host computation, clock-reference generation, and optical-pump generation are outside 4.2 K; their conducted and absorbed heat is still charged at the cold stage. The always-on safety interlock must fit within the overhead. Burst scheduling must demonstrate transient temperatures, wake-up settling, readout/QEC availability, and the stated gate and recovery latencies; 25% activity is not continuous full-array throughput. Continuous operation of all four arrays at 4.2 K is rejected under this budget. Moving blocks to a warmer stage is an alternative requiring a new wiring, noise, and conduction budget, not an implicit escape from the accounting. The FD-SOI model uses threshold shifts approximately +180 mV (n) and −210 mV (p), a forward well bias 1.2 V, back-gate sensitivity 85.4 mV/V, and threshold spread 1.42 mV corrected by four-bit trim DACs. These are design inputs requiring device-level calibration.

4.5 Heterodyne readout and clock specification

The local oscillator is +3.50 dBm with RIN −165.0 dBc/Hz; the InGaAs detectors have nominal efficiency 98.51% and dark current 12.5 pA.

Readout quantity	Nominal value
Signal current	12.40 μ A
LO shot-noise current density	4.12×10^{-12} A/ $\sqrt{\text{Hz}}$
Amplifier input-noise density	1.05×10^{-12} A/ $\sqrt{\text{Hz}}$
LO RIN current-noise density	0.42×10^{-12} A/ $\sqrt{\text{Hz}}$
Demodulation bandwidth	83.33 MHz
Integrated crosstalk	−42.50 dB
Receiver SNR	18.42 dB
Single-shot fidelity (12.0 ns)	99.9821%

For independent input-current noise, $\sigma_I^2 = \int S_I(f) |H(f)|^2 df$. A fidelity prediction shall additionally specify the state-dependent current separation, priors, threshold, detector response, and non-Gaussian tails. V-31 acceptance shall verify the predicted fidelity through calibrated single-shot classification. V-30 acceptance shall derive DAC ENOB from measured SINAD; SFDR shall be characterized separately. The 40.000 GHz sapphire-referenced DRO phase-noise record is:

Offset	Nominal $L(f)$ (dBc/Hz)	Ceiling (dBc/Hz)
10 Hz	−45.2	−40.0
100 Hz	−78.4	−72.0
1 kHz	−108.6	−102.0
10 kHz	−128.2	−124.0
100 kHz	−138.5	−134.0
1 MHz	−148.1	−144.0
10 MHz	−156.4	−152.0

For single-sideband phase noise,

$$\sigma_t = \frac{1}{2\pi f_0} \sqrt{2 \int_{10 \text{ Hz}}^{10 \text{ MHz}} 10^{L(f)/10} df}.$$

The nominal jitter is 6.42 fs and Allan deviation is $\sigma_y(1 \text{ s}) = 8.15 \times 10^{-14}$ against 1.20×10^{-13} . An integration/interpolation convention is needed to reproduce jitter from only seven samples; these points alone do not independently validate 6.42 fs.

5 Holographic reconstruction and geometric constraints

5.1 Discrete HKLL reconstruction

HKLL constructs semiclassical bulk operators from boundary observables under an AdS/CFT dictionary [4]. A proposed discrete emulator uses

$$\phi_{\text{bulk}}(z, x, t) = \sum_{k=1}^{N_{\text{ch}}} \int d\tau K_{\text{HKLL}}(z, x, t | y_k, \tau) \mathcal{O}_k(\tau).$$

Cell weights must be absorbed into the kernel when $\mathcal{O}_k$ is a cell average. The boundary conditions, scaling dimension, normalization, temporal support, lattice quadrature rule, and convergence study are not specified by writing this expression. Bath memory modifies the dynamical reconstruction problem but does not by itself derive a specific encoding defect or a quantum error-correction threshold.

Definition 5.1 (Domain-consistent approximate isometry). *Let P_{code} act on a bulk domain $\mathcal{H}_B$ and let $W : \mathcal{H}_B \rightarrow \mathcal{H}_{\text{phys}}$ satisfy $W = WP_{\text{code}}$. Define*

$$\varepsilon_W = \|W^\dagger W - P_{\text{code}}\|_{\text{op}}.$$

If $\mathcal{H}_B$ is already the code space, $P_{\text{code}} = I_B$. The boundary range projector is not $W^\dagger W$; for an approximate isometry it is not generally $WW^\dagger$ either.

For $\varepsilon_W < 1$, code-vector norms obey $(1 - \varepsilon_W)\|v\|^2 \leq \|Wv\|^2 \leq (1 + \varepsilon_W)\|v\|^2$. This follows by applying the operator-norm bound to $\langle v, (W^\dagger W - I)v \rangle$. It does not establish correctability of an unspecified noise channel. The exact Knill–Laflamme condition $PE_i^\dagger E_j P = \lambda_{ij} P$ needs actual error operators; an approximate condition also needs a defined recovery map and operational error norm.

5.2 V-05 isometry-defect standard and model evaluation

The physical encoding shall satisfy the primary V-05 design and verification standard

$$\varepsilon_W := \|W^\dagger W - P_{\text{code}}\|_{\text{op}} \leq 3.430 \times 10^{-3}. \quad (24)$$

The channel-count fit used for design evaluation is

$$\varepsilon_W^{\text{fit}}(N_{\text{ch}}) = (3.15 \times 10^{-4}) N_{\text{ch}}^{0.395} + 3.82 \times 10^{-4}. \quad (25)$$

Here N_{ch} is the channel count, distinct from the parent affine level. Direct evaluation gives $\varepsilon_W^{\text{fit}}(312) = 3.4263993 \times 10^{-3}$, below the V-05 ceiling 3.430×10^{-3} by 3.6007×10^{-6} . Verification shall record this result as a model evaluation. Physical acceptance shall require the measured defect plus its stated uncertainty to meet the same ceiling.

N_{ch}	Nominal estimate	Evaluated fit	Reconstruction error	$\Delta Z/Z$	z_{max}/R
6	1.02×10^{-3}	1.02126×10^{-3}	4.85×10^{-3}	2.84×10^{-6}	0.182
26	1.52×10^{-3}	1.52284×10^{-3}	1.94×10^{-3}	8.12×10^{-7}	0.415
104	2.33×10^{-3}	2.35460×10^{-3}	5.12×10^{-4}	2.95×10^{-7}	0.784
312	3.42640×10^{-3}	3.42640×10^{-3}	1.45×10^{-4}	1.18×10^{-7}	0.985

All tabulated values are model estimates; the 312-channel estimate uses the evaluated fit. Physical acceptance shall independently verify the V-05 defect and the V-03 completed-partition modular residual using their specified observables and protocols.

Proposition 5.2 (Unitary remapping does not repair a Gram defect). *If U is a unitary boundary transformation, then $(UW)^\dagger(UW) = W^\dagger W$. Consequently a column permutation or Givens rotation implemented as U_{iso} cannot decrease $\|W^\dagger W - I\|_{\text{op}}$. For W stored with boundary rows and code columns, a column operation is instead $W \mapsto WV$ with V unitary on the code domain; then $(WV)^\dagger(WV) - I = V^\dagger(W^\dagger W - I)V$ has the same operator norm and spectrum. For a larger domain, the code projector must transform with the code basis.*

Proof. Substitute $U^\dagger U = I$; for code-basis rotations use unitary invariance of the operator norm. $\square$

Reconstruction after an erasure requires an actual recovery or recalibrated encoding, not merely a norm-preserving software rotation. Nonunitary channel recalibration or explicit recovery projections (with success probability and leakage accounted for) are required. Each updated encoding shall meet Equation 24, including its calibration uncertainty. On the code domain, polar normalization $\widetilde{W} = W(W^\dagger W)^{-1/2}$ gives an exact mathematical isometry when the Gram matrix is positive definite, but its physical implementation and noise cost are additional problems. A scalar energy-conservation test is not an operator-norm certificate.

5.3 Trace-free excitations and the electric Weyl tensor

For finite-dimensional excitation matrices, off-diagonal hopping combinations have $\text{Tr } \hat{O}_{\text{exc}} = 0$. This Hilbert-space trace is not a spacetime contraction $T^\mu{}_\mu$, and neither implies vanishing traceless stress. For example, $\text{diag}(1, -1)$ has zero trace but nonzero anisotropy and a nonzero expectation in its first eigenstate. Thus the requested general implication $\text{Tr } \hat{O}_{\text{exc}} = 0 \Rightarrow E_{\mu\nu} = 0$ does not follow from the stated premise.

Theorem 5.3 (Vanishing electric Weyl tensor in the stated AdS_3 bulk). *For a three-dimensional metric, the Weyl tensor vanishes identically. In particular, $E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta = 0$ for any normal n , regardless of whether excitation operators have zero trace.*

Proof. In three dimensions the algebraic curvature decomposition is

$$R_{abcd} = g_{ac}R_{bd} - g_{ad}R_{bc} - g_{bc}R_{ad} + g_{bd}R_{ac} \\ - \frac{R}{2}(g_{ac}g_{bd} - g_{ad}g_{bc}).$$

The Weyl tensor is the remainder after subtracting exactly these Ricci terms; therefore $C_{abcd} = 0$, and its contraction with two normals is zero [5]. $\square$

This supplies the zero-Weyl result for the specified metric $ds^2 = R^2 z^{-2}(dz^2 - dt^2 + dx^2)$. It does not imply an absence of all tidal curvature, conformal flatness of every three-dimensional metric (the Cotton tensor is relevant there), or error correction. In four or more dimensions, vacuum solutions can have nonzero Weyl curvature despite a trace-free or vanishing stress tensor. A higher-dimensional extension needs independent conformal-flatness or curvature constraints; it cannot reuse an invalid trace argument. This AdS_3 emulator theorem does not establish higher-dimensional cosmological closure from the affine-level locks alone.

6 Thermodynamics and SK-9 compilation

6.1 Landauer bound and why saturation is not demonstrated

For an initially uncorrelated system and thermal reservoir, a joint unitary erasure process obeys the entropy-balance identity [6]

$$\frac{Q}{k_B T} = \Delta S + I(S' : R') + D(\rho'_R \| \rho_R),$$

where entropies are in nats and $\Delta S = S(\rho_S) - S(\rho'_S)$. The last two terms are nonnegative; resetting an initially unbiased bit completely gives $\Delta S = \ln 2$ and $Q \geq k_B T \ln 2$. Equality requires the reversible limiting conditions, not merely a low-energy transistor design. Biased bits, incomplete reset, and side information require the appropriate entropy decrease rather than an unconditional one-bit count. At the specified temperature,

$$E_L = (1.380649 \times 10^{-23})(4.2) \ln 2 = 4.019370439 \times 10^{-23} \text{ J/bit} \simeq 4.0194 \times 10^{-23} \text{ J/bit}.$$

The requested physical design value $E_{\text{phys}} = 8.14 \times 10^{-18} \text{ J/bit}$ is approximately 50.81 eV and 2.0252×10^5 times this floor. It is *not* saturation per erased bit. An assertion that each transition erases that many independent bits needs a demonstrated entropy-flow accounting and cannot be inferred from this ratio.

For the 2EE2P adiabatic transition model,

$$\begin{aligned} E_{\text{diss}} &= 2(R_{\text{on}}C/\tau_r)CV_{dd}^2 + \frac{1}{2}CV_{th}^2 + I_{\text{leak}}V_{dd}\tau_r, \\ R_{\text{on}} &= 1.20 \text{ k}\Omega, \quad C = 14.20 \text{ fF}, \quad \tau_r = 420.0 \text{ ps}, \quad V_{dd} = 0.65 \text{ V}, \\ V_{th} &= 0.28 \text{ V}, \quad I_{\text{leak}} = 4.12 \text{ nA}. \end{aligned} \tag{26}$$

Direct substitution gives

$$E_{\text{diss}} = \underbrace{4.8682 \times 10^{-16}}_{\text{dynamic 1}} + \underbrace{5.5664 \times 10^{-16}}_{\text{dynamic 2}} + \underbrace{1.1248 \times 10^{-18}}_{\text{leakage}} \simeq 1.0446 \times 10^{-15} \text{ J}.$$

The resulting dissipation exceeds the $8.14 \times 10^{-18} \text{ J}$ transition-energy target. Adding passive interconnect losses with $R_{\text{trace}} = 8.5 \Omega$, $C_{\text{trace}} = 180 \text{ fF}$ cannot reduce that total. Energy recovery would need an explicit waveform and recovered-work term. Also, the stated line count and activity rate give

$$P_{\text{dyn}} = 11776(8.14 \times 10^{-18})(8.7719 \times 10^8) = 8.40845 \times 10^{-5} \text{ W} = 0.0840845 \text{ mW},$$

not 84.20 mW. The discrepancy is approximately a factor of 1000, so the 84.20 mW thermal allocation cannot be derived from this transition-energy model. Physical saturation is not established. The duty-cycled thermal allocation is a testable operating contract, not a measurement validating the transition-energy model.

6.2 Depth-9 Solovay–Kitaev: conditional arithmetic versus a compiler

The compiler design specifies a 124-word braid library. Its implementation shall define the matrices, anyon representation, enumerated words, and certified covering net. A list cardinality alone does not define a universal gate set. For the standard recursive construction [7], suppose the required inverse-closed dense gate set and certified base net exist, with

$$\varepsilon_n \leq c_{\text{SK}} \varepsilon_{n-1}^{3/2}, \quad \ell_n \leq 5\ell_{n-1}, \quad c_{\text{SK}} = 4, \quad \varepsilon_0 = 6.25 \times 10^{-8}.$$

Set $x_n = c_{\text{SK}}^2 \varepsilon_n$ and iterate $x_n \leq x_{n-1}^{3/2}$ to obtain

$$\varepsilon_9 \leq \frac{(16\varepsilon_0)^{(3/2)^9}}{16} = \frac{10^{-230.66015625}}{16} \simeq 1.37 \times 10^{-232} < 10^{-230}.$$

This is a *conditional ideal approximation bound*, not a demonstrated hardware error floor. A 124-element library cannot itself be a 6.25×10^{-8} covering net of all $SU(2)$: locally the three-dimensional group requires a number of radius- ε balls scaling as ε^{-3} . The 124 words may instead generate much longer base-net sequences, whose search procedure and coverage shall be specified and certified.

The word-length multiplier is $5^9 = 1953125$. If $\ell_0 = 3$, there can be 5859375 elementary gates, taking 83.26 ms at 14.21 ns per gate. Caching compilation removes classical search time, not

physical gate execution or accumulated noise. Storing 124 such words at one byte per elementary symbol would require approximately 727 MB, not a 1472-byte ledger. The implementation stores 124 compact 8-byte generator/recursion descriptors (992 bytes) in the 1472-byte ledger, leaving 480 bytes for syndrome/ECC/control metadata. Each descriptor identifies a separately specified immutable generator table, a base-word index, recursion depth, flags, and a sequence identifier; it is not a compressed arbitrary million-gate string. A deterministic expansion recipe and target-specific recursion data must be external or streamed and explicitly budgeted. Neither 727 MB of expanded storage nor 10^{-230} -precision target matrices are claimed to fit in this arena. Certification to 10^{-230} requires substantially more than binary64 precision (roughly 765 binary precision bits before guard digits). Compiler qualification shall provide a numerical certificate at the required precision.

7 Quantum error correction and digital-twin specification

7.1 Threshold and decoder model parameters

Physical error rate	Logical error rate	Leakage	Syndrome success
2.50×10^{-2}	1.42×10^{-1}	3.12×10^{-3}	0.742
1.84×10^{-2}	1.84×10^{-2}	8.45×10^{-4}	0.945
1.00×10^{-2}	3.20×10^{-3}	2.11×10^{-4}	0.988
1.20×10^{-3}	1.45×10^{-5}	4.12×10^{-5}	0.99982
5.00×10^{-4}	2.10×10^{-7}	8.50×10^{-6}	0.99998

The second row is labeled the threshold point and the fourth the operating point. A crossing of physical and logical error rates in one finite system does not alone establish a fault-tolerance threshold: code families, circuit noise, decoder, sample size, and finite-size scaling are needed.

MPS bond χ	Latency (μ s)	Energy/syndrome (nJ)	Frame-drop rate	Residual entropy (bits)
64	38.2	8.4	0	0.142
128	74.5	18.2	0	0.038
256	142.8	42.1	0	0.0012
512	312.4	98.6	4.12×10^{-6}	0.0001

The decoder targets 128 parallel VPUs and 14.3 TB/s aggregate dual-port SRAM bandwidth. The $\chi = 256$ operating point is below V-14's 150 μ s ceiling. The complete pipeline budget is 12.0 ns ingestion plus 142.8 μ s decoding plus 18.5 ns command issue, totaling 142.8305 μ s; the nominal dispatch jitter is at most 4.2 ps. The pipeline latency is separate from the V-34 ECC maximum 1.20 ns and the V-27 recovery maximum 120.00 ns.

7.2 Multiphysics engine: equations and numerical acceptance

The digital twin assigns `simulator/multi_physics/thermal_fea_sp_sim.py` to thermal, vibration, and interposer models. Its thermal class uses a $20 \times 20 \times 10$ grid, $\Delta x = 1 \mu\text{m}$, $\Delta t = 10 \text{ ps}$, $\rho_{\text{SOI}} = 2330$ and $\rho_{\text{InP}} = 4810 \text{ kg/m}^3$, and the following approximate laws:

$$\begin{aligned} c_{p,\text{SOI}} &= 1.6 \times 10^{-3} T^3 + 10^{-6}, & c_{p,\text{InP}} &= 2.1 \times 10^{-3} T^3 + 10^{-6} \quad [\text{J}/(\text{kg K})], \\ \kappa_{\text{SOI}} &= 1.2 \times 10^{-4} T^3 + 10^{-5}, & \kappa_{\text{InP}} &= 8.5 \times 10^{-4} T^3 + 10^{-5} \quad [\text{W}/(\text{m K})]. \end{aligned}$$

Its nominal quench is 10^{14} W/m^3 in index region (8:12, 8:12, 0:2). The continuum equation is

$$\rho c_p(T) \partial_t T = \nabla \cdot [\kappa(T) \nabla T] + \dot{q}.$$

A conservative finite-volume discretization assigns one interface flux to each face and equal/opposite energy transfers to its neighbors. Replacing the divergence by $\kappa \nabla^2 T$ discards $\nabla \kappa \cdot \nabla T$ when conductivity varies. For constant diffusivity α , a three-dimensional explicit central scheme requires

$\Delta t \leq \Delta x^2/(6\alpha)$; interface conductance adds a cell-capacity constraint. Acceptance requires energy conservation, nonnegative temperatures, mesh/time refinement, boundary-condition checks, and independent thermal-contact calibration.

Material arrays shall use the same index domain as each flux or temperature update. At a thermal contact, the Kapitza flux shall replace the bulk face flux so that each interface transfer is counted once. Each simulation shall record the mesh, material parameters, boundary conditions, heat-source region, and solver tolerances.

The interposer model uses propagation/skin-effect estimates and an RLC PDN equivalent circuit; the vibration model uses a calibrated transfer function. Three-dimensional finite-element qualification additionally requires mesh and solver convergence. The module interfaces and integration requirements are specified in Appendix C.

7.3 CAD, yield, and coupling-model contract

The `simulator/cad_yield/spc_photonic_cad.py` model shall draw fabrication tolerances from the measured process means, standard deviations, and correlations. Nominal operating values are aerogel index 1.0182, sidewall angle 89.88 degrees, and waveguide critical dimension 250.12 nm. A local effective-index model may use

$$n_{\text{eff}} = n_{\text{eff},0} + s_n(n_a - n_{a,0}) + s_\theta(\theta - \theta_0) + s_w(w - w_0),$$

where all sensitivities, units, and the reference index shall be calibrated for the specified geometry. Resonance predictions shall use the calibrated optical length and mode order. Each run shall record its random seed, acceptance windows, trimming operations, and spare-assignment policy. V-35 evaluates the resulting at-least-312-of-336 functional-channel module yield, not a single process-step yield.

For the negative-binomial defect model,

$$Y_{\text{step}} = (1 + D_0 A / \alpha_c)^{-\alpha_c}, \quad D_0 = 0.12 \text{ cm}^{-2}, \quad A = 0.045 \text{ mm}^2 = 4.5 \times 10^{-4} \text{ cm}^2, \quad \alpha_c = 1.85.$$

This gives approximately 99.9946% for those particular parameters, not the entire process yield. The separate six-step yield budget is:

Process	Step yield (%)	Cumulative (%)
Quantum-well epitaxy	99.45	99.45
Electron-beam patterning	98.80	98.26
ICP-RIE	97.90	96.20
Sol-gel deposition	98.15	94.42
Supercritical drying	98.70	93.19
Flip-chip bonding	94.88	88.42

Module acceptance is at least 312 functional channels out of 336, a within-module fraction of 92.857%; V-35's 85% is a *module manufacturing yield*, not the same fraction. For independent channel survival probability p , module yield would be $\sum_{j=312}^{336} \binom{336}{j} p^j (1-p)^{336-j}$; clustered failures require a joint model.

EDA release artifacts Wafer-ready mask layouts are generated by `eda/exporters/pdk_hexamer_exporter.py` (gdsfactory GDSII for C_6 -symmetric hexamers, directional couplers, and thermo-optic phase shifters) and the 12-layer Rogers RO4350B interposer S-parameters are exported by `eda/exporters/export_interposer_rf.py` as a 2-port Touchstone `.s2p` file up to 40 GHz with $Z_0 = 50.12 \Omega$ reference.

7.4 HIL bus, MPS state, and formal-release contract

The HIL module `simulator/hil_qec/hil_mps_qec.py` shall implement a virtual MMIO bus, an MPS state model, error injection, and a closed-loop recovery trace. Firmware and HIL shall use SHBT-MMIO-1 in Section 8.2, including identical status bits, channel snapshots, command-completion semantics, and fault-latch behavior. The virtual bus shall model device side effects as well as register storage. The MPS is

$$|\psi\rangle = \sum_{s_1 \dots s_N} A_1^{s_1} A_2^{s_2} \dots A_N^{s_N} |s_1 \dots s_N\rangle,$$

with compatible bond dimensions, normalization, and contractions that preserve boundary conditions. Pauli injection, stabilizer measurement, recovery selection, and state overlap are different operations. A toy marginal or tensor norm is not a syndrome decoder or a heterodyne readout fidelity. The acceptance suite needs exact small-system comparisons, truncation-error bounds, reproducible correlated-noise seeds, logical observables, and confidence intervals.

The implementation release chain is `formal/formal_verification.py` (Z3 invariant proofs), `eda/exporters/pdk_hexamer_exporter.py` (gdsfactory GDSII masks), `eda/exporters/export_interposer_rf.py` (2-port Touchstone S2P), `cli/main.py` (unifying `shbt-tool` CLI), and `tests/run_all_tests.py` (master end-to-end runner). The `build-kernel` subcommand compiles `shbt-reference.so`, `sim` runs the multi-physics and MPS-QEC co-simulation, `export-eda` generates GDSII and S2P artifacts, and `verify` runs the Z3 proofs and `pytest` closed-loop bench. A conforming release must reject SMT `sat` and `unknown` when checking a forbidden behavior; only `unsat` discharges that particular model query [10]. A mock ELF or a compiler-produced relocatable object is not a bootable, linked microkernel. Release qualification requires a linker script, startup code, entry point, relocation verification, ISA checks, binary-level WCET evidence, and hardware binding to the verified register map.

8 Bare-metal shbt-os memory, interfaces, and reference C

8.1 The 2112-byte UnifiedStinespringFrame

The specified zero-heap working arena is 64-byte aligned and contains 33 cache lines:

$$S = 2112 \text{ B}, \quad S_A = 640 \text{ B} = 10 \times 64 \text{ B}, \quad S_D = 1472 \text{ B} = 23 \times 64 \text{ B}, \\ \eta_A = 10/33, \quad \eta_D = 23/33.$$

The Active Visible Register has 160 binary32 entries; the Dark Ledger has 1472 bytes for syndrome records, ECC metadata, and compact command descriptors. These are engineering allocations, not a theorem derived from the canonical affine-level vector. A Stinespring channel $\mathcal{E}(\rho) = \text{Tr}_E(V\rho V^\dagger)$ has an environment dimension determined by its Kraus rank, at most $d_{\text{in}}d_{\text{out}}$ for a finite channel, not the asserted “57 real parameters per node”: substituting the affine level k_q into a finite-node dimension formula is unjustified. A classical byte array is not a quantum environment and does not represent the completed dark CFT Hilbert space.

The 640-byte region cannot store 312 complex binary32 amplitudes: those require $312 \times 2 \times 4 = 2496$ bytes, or 2560 bytes if each plane is padded to 320 elements. Streaming tile buffers, external static matrices, and stacks must therefore be budgeted separately. A zero heap is not a zero stack. Protecting 2112 payload bytes using one check byte per 64-bit data word adds 264 ECC bytes unless the memory controller stores them out of band. Build-time C layout assertions shall verify the payload size and alignment.

8.2 SHBT-MMIO-1 register standard

Firmware, HDL, and digital-twin simulation shall implement **SHBT-MMIO-1** as the primary register-interface design and verification standard. The register bank comprises fourteen

aligned 32-bit words at base 0x70000000, with word offsets 0x00–0x34 and byte extent 0x70000000–0x70000037 (56 bytes). Platform binding shall require ABI version 1 before issuing control commands.

Offset	Register	Required access and behavior
0x00	<code>status</code>	Read-only: bit 0 <code>OVERTEMP</code> , bit 1 <code>PLL_LOCK</code> , bit 2 <code>ECC_ERR</code> , bit 3 <code>FAULT_ST</code>
0x04	<code>blank</code>	Read/write: 1 blanks all active drive channels; 0 requests enable (only when healthy)
0x08	<code>fifo_data</code>	Bit 0 flush request, bit 1 busy; requests self-clear in hardware
0x0C	<code>pll_ctrl</code>	Write 1 requests lock; <code>status</code> bit 1 indicates completion
0x10	<code>ecc_low</code>	Read/write: low 32 bits of one latched payload word
0x14	<code>ecc_high</code>	Read/write: high 32 bits of one latched payload word
0x18	<code>ecc_check</code>	Read/write: bits 0–6 are Hamming parity; bit 7 is overall even parity
0x1C	<code>ecc_commit</code>	Write 1 commits the corrected payload and check byte atomically
0x20	<code>fault_latch</code>	Write-one-to-set software fault; only a separate authorized reset clears
0x24	<code>channel_select</code>	Read/write: channel 0–311; completed write latches its coherent ECC word
0x28	<code>abi_version</code>	Read-only: <code>SHBT_MMIO_ABI_VERSION == 1</code> ; platform binding shall reject any other version
0x2C	<code>phase_offset</code>	Read/write: 32-bit phase-calibration payload; scaling set by calibration record
0x30	<code>ecc_counts</code>	Read-only: corrected count bits 0–7, uncorrectable count bits 8–15; saturating
0x34	<code>control</code>	Write-one requests: start bit 0, soft reset bit 1, recovery bit 2; pump-attenuation payload bits 8–15; reserved bits zero

Writes shall be strongly ordered device accesses with a completion read. Hardware shall provide a coherent ECC snapshot until commit, uncached MMIO mappings, autonomous overtemperature blanking, and a healthy-status gate on every enable request. Platform binding shall establish these bus guarantees independently of the C `volatile` qualifier. The reference implementation specifies bounded polling, zero-heap operation, architecture-appropriate ordering, and explicit failure returns for uncorrectable ECC or persistent faults. Target qualification shall verify worst-case execution time separately from functional correctness. Every command register shall support a side-effect-free completion read; readback shall not reissue its command. Reset requests shall preserve unsafe interlocks and the independently authorized fault latch.

Status and command semantics. `OVERTEMP` is active-high unsafe; `PLL_LOCK` is active-high locked at status bit 1. Status bits 2 and 3 identify ECC error and fault state respectively. Software shall not write `status` or `ecc_counts`. ECC commit and device state transitions update the status flags; writing zero to `fault_latch` shall not clear it. Reset and recovery commands shall preserve independent interlocks. Additional telemetry shall use a versioned extension rather than aliasing these offsets.

8.3 SECDED Hamming(72,64) ECC standard

SECDED Hamming(72,64) ECC is the primary payload-protection design and verification standard. Each protected 64-bit payload shall use this code. Positions 1–71 contain seven Hamming parity positions 1, 2, 4, 8, 16, 32, 64 and 64 data positions. Check-byte bits 0–6 store those seven parity bits; bit 7 is an overall even-parity bit over the other 71 bits. Let s be the XOR of occupied Hamming positions and p the overall parity mismatch. The complete decision table is:

<i>s</i>	<i>p</i>	Decision under at most two bit errors
0	0	Clean codeword
0	1	Correct overall check bit
1...71	1	Correct indicated data or Hamming parity position
nonzero	0	Detected double-bit error; do not modify payload
72...127	1	Invalid shortened-code syndrome; report uncorrectable

An error in one data bit changes *s* to that bit's code position and flips *p*; two distinct errors give even *p* and nonzero *s*. This proves single-error correction and double-error detection for the stated fault model. Three or more errors may be miscorrected; SECDDED is not a general integrity checksum. Implementations shall use codeword-position syndromes and overall even parity. V-34 requires a decode time $t_{\text{ecc}} \leq 1.20 \text{ ns}$; functional correction tests and target timing tests shall be evaluated separately.

8.4 Complete reference recovery and ECC translation unit

The following freestanding C11 reference listing specifies the runtime interface for `kernel/src/shbt_core_runtime.c`. It uses the register layout in `kernel/include/shbt_hardware.h` (Section 8.2); target qualification shall establish its complete instruction and device-access timing.

Listing 1: SHBT-R freestanding microkernel source (`kernel/src/shbt_core_runtime.c`).

```

/*
 * shbt_core_runtime.c - SHBT-R bare-metal microkernel core.
 *
 * Freestanding C11. No standard C runtime is used: there is no heap, no
 * libc, and no hosted assumptions. The only headers are compiler-provided
 * (<stdint.h>, <stddef.h>, <immintrin.h> for AVX-512) and the hardware MMIO
 * definition in kernel/include/shbt_hardware.h.
 */

#include <stdint.h>
#include <stddef.h>
#include <immintrin.h>
#include "shbt_hardware.h"

/* -----
 * Platform timing (x86 TSC; fallback is intentionally empty for non-x86)
 * ----- */
#ifndef SHBT_TSC_HZ
#define SHBT_TSC_HZ 3000000000ULL /* 3.0 GHz default; override at build time */
#endif

static inline uint64_t shbt_cycles(void)
{
    #if defined(__x86_64__) || defined(__i386__)
        uint32_t lo, hi;
        __asm__ __volatile__ ("lfence\n\t; rdtsc" : "=a"(lo), "=d"(hi) : "memory");
        return ((uint64_t)hi << 32) | lo;
    #else
        return 0ULL;
    #endif
}

/* -----
 * SECDDED Hamming(72,64) ECC
 * ----- */

/* Data bits covered by each of the 7 Hamming check bits. */
static const uint64_t shbt_ecc_mask[7] = {
    UINT64_C(0xab555555556aaad5b),
    UINT64_C(0xcd99999999b33366d),

```

```

    UINT64_C(0xf1e1e1e1e3c3c78e),
    UINT64_C(0x01fe01fe03fc07f0),
    UINT64_C(0x001fffe0003fff800),
    UINT64_C(0x001ffffffffc000000),
    UINT64_C(0xfe00000000000000),
};

/* Map Hamming syndrome (codeword position) to data bit index, or -1. */
static const int8_t shbt_ecc_data_of_pos[128] = {
    -1, -1, -1, 0, -1, 1, 2, 3,
    -1, 4, 5, 6, 7, 8, 9, 10,
    -1, 11, 12, 13, 14, 15, 16, 17,
    18, 19, 20, 21, 22, 23, 24, 25,
    -1, 26, 27, 28, 29, 30, 31, 32,
    33, 34, 35, 36, 37, 38, 39, 40,
    41, 42, 43, 44, 45, 46, 47, 48,
    49, 50, 51, 52, 53, 54, 55, 56,
    -1, 57, 58, 59, 60, 61, 62, 63,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
    -1, -1, -1, -1, -1, -1, -1, -1,
};

typedef struct {
    uint64_t corrected_data;
    uint8_t corrected_check;
    uint8_t single_bit_error;
    uint8_t double_bit_error;
} secdec_result_t;

static inline uint8_t shbt_ecc_hamming_bits(uint64_t data)
{
    uint8_t c = 0;
    for (int j = 0; j < 7; ++j)
        c |= (uint8_t)(__builtin_parityll(data & shbt_ecc_mask[j]) << j);
    return c;
}

uint8_t shbt_ecc_encode(uint64_t data)
{
    uint8_t c = shbt_ecc_hamming_bits(data);
    uint8_t overall = (uint8_t)(__builtin_parityll(data) ^ __builtin_parity(c));
    return (uint8_t)(c | (overall << 7));
}

secdec_result_t shbt_ecc_decode(uint64_t data, uint8_t check_code)
{
    uint8_t expected = shbt_ecc_hamming_bits(data);
    uint8_t syn = (uint8_t)((check_code ^ expected) & 0x7FU);
    uint8_t parity = (uint8_t)(__builtin_parityll(data) ^ __builtin_parity(check_code));

    secdec_result_t r = { data, check_code, 0, 0 };

    if (syn == 0 && parity == 0)
        return r;

    if (syn == 0 && parity == 1) {
        /* Single-bit error in the overall parity bit. */
        r.corrected_check ^= 0x80U;
        r.single_bit_error = 1;
    }
}

```

```

        return r;
    }

    if (parity == 0) {
        /* Even number of bit flips (detected but uncorrectable). */
        r.double_bit_error = 1;
        return r;
    }

    /* Single-bit error at codeword position 'syn'. */
    if ((syn & (syn - 1)) == 0) {
        /* Check-bit position (power of two in 1..64). */
        int j = __builtin_ctz((unsigned int)syn);
        r.corrected_check ^= (uint8_t)(1U << j);
        r.single_bit_error = 1;
        return r;
    }

    int8_t db = shbt_ecc_data_of_pos[syn];
    if (db < 0) {
        /* Syndrome points outside the valid 72-bit codeword. */
        r.double_bit_error = 1;
        return r;
    }

    r.corrected_data ^= (UINT64_C(1) << db);
    r.single_bit_error = 1;
    return r;
}

/* -----
 * Post-quench recovery driver
 * ----- */

#define SHBT_RECOVERY_BUDGET_NS 120ULL
#define SHBT_RECOVERY_PLL_WAIT_ITERS 1000000U

static inline int shbt_recovery_expired(uint64_t started)
{
    uint64_t budget = (SHBT_TSC_HZ * SHBT_RECOVERY_BUDGET_NS)
        / 1000000000ULL;
    return shbt_cycles() - started > budget;
}

static int32_t shbt_fail_closed(ShbtRegisters *hardware, int32_t result)
{
    hardware->blank = 1U;
    (void)hardware->blank;
    hardware->fault_latch = 1U;
    (void)hardware->fault_latch;
    return result;
}

int32_t shbt_recover(void)
{
    ShbtRegisters *hardware = SHBT_MMIO;
    uint64_t started = shbt_cycles();
    uint32_t unsafe = SHBT_STATUS_OVERTEMP | SHBT_STATUS_FAULT_ST;

    if (hardware->abi_version != 1U)
        return -3;
    if ((hardware->status & unsafe) || hardware->fault_latch)
        return shbt_fail_closed(hardware, -1);
}

```

```

hardware->blank = 1U;
(void)hardware->blank;
if (shbt_recovery_expired(started))
    return shbt_fail_closed(hardware, -2);

hardware->fifo_data = 1U;
(void)hardware->fifo_data;
uint32_t polls = 0;
while (hardware->fifo_data & 2U) {
    if ((hardware->status & unsafe) || hardware->fault_latch)
        return shbt_fail_closed(hardware, -1);
    if (++polls >= SHBT_RECOVERY_PLL_WAIT_ITERS
        || shbt_recovery_expired(started))
        return shbt_fail_closed(hardware, -2);
}

uint32_t channel = hardware->channel_select;
if (channel >= 312U)
    return shbt_fail_closed(hardware, -3);
hardware->channel_select = channel;
(void)hardware->channel_select;
uint64_t data = ((uint64_t)hardware->ecc_high << 32)
    | hardware->ecc_low;
secded_result_t decoded = shbt_ecc_decode(data, (uint8_t)hardware->ecc_check);
if (decoded.double_bit_error)
    return shbt_fail_closed(hardware, -1);
if (shbt_recovery_expired(started))
    return shbt_fail_closed(hardware, -2);
if (decoded.single_bit_error) {
    hardware->ecc_low = (uint32_t)decoded.corrected_data;
    (void)hardware->ecc_low;
    hardware->ecc_high = (uint32_t)(decoded.corrected_data >> 32);
    (void)hardware->ecc_high;
    hardware->ecc_check = decoded.corrected_check;
    (void)hardware->ecc_check;
    hardware->ecc_commit = 1U;
    (void)hardware->ecc_commit;
}

hardware->pll_ctrl = 1U;
(void)hardware->pll_ctrl;
polls = 0;
while (!(hardware->status & SHBT_STATUS_PLL_LOCK)) {
    if ((hardware->status & unsafe) || hardware->fault_latch)
        return shbt_fail_closed(hardware, -1);
    if (++polls >= SHBT_RECOVERY_PLL_WAIT_ITERS
        || shbt_recovery_expired(started))
        return shbt_fail_closed(hardware, -2);
}
if ((hardware->status & (unsafe | SHBT_STATUS_ECC_ERR))
    || hardware->fault_latch)
    return shbt_fail_closed(hardware, -1);
if (shbt_recovery_expired(started))
    return shbt_fail_closed(hardware, -2);
hardware->blank = 0U;
if (hardware->blank || hardware->fault_latch
    || (hardware->status & (unsafe | SHBT_STATUS_ECC_ERR))
    || !(hardware->status & SHBT_STATUS_PLL_LOCK))
    return shbt_fail_closed(hardware, -1);
if (shbt_recovery_expired(started))
    return shbt_fail_closed(hardware, -2);
return 0;
}

```

```

/* -----
 * AVX-512 Givens rotation column remapping
 * ----- */

void shbt_remap(double *col_a, double *col_b, double c, double s, size_t n)
{
#ifdef __AVX512F__
    __m512d vc = _mm512_set1_pd(c);
    __m512d vs = _mm512_set1_pd(s);

    for (; n >= 8; n -= 8, col_a += 8, col_b += 8) {
        __m512d a = _mm512_loadu_pd(col_a);
        __m512d b = _mm512_loadu_pd(col_b);

        __m512d an = _mm512_add_pd(_mm512_mul_pd(vc, a), _mm512_mul_pd(vs, b));
        __m512d bn = _mm512_sub_pd(_mm512_mul_pd(vc, b), _mm512_mul_pd(vs, a));

        _mm512_storeu_pd(col_a, an);
        _mm512_storeu_pd(col_b, bn);
    }
#endif

    for (size_t i = 0; i < n; ++i) {
        double a = col_a[i];
        double b = col_b[i];
        col_a[i] = c * a + s * b;
        col_b[i] = c * b - s * a;
    }
}

```

The platform verifies ABI version 1 before enabling interrupts; channel selection latches the coherent word until commit, as required by the unified map. Descriptor bytes must be copied with a defined serialization/endian convention rather than casting potentially unaligned external buffers to C structs. An asynchronous fault may arrive after the final software status read, so the specified hardware healthy-gate must reject an unsafe enable independently of software. This is an essential part of the interface contract, not something a C unit test can prove.

8.5 Complete AVX-512 remapping implementation

The AVX-512 Givens rotation remap in `kernel/src/shbt_core_runtime.c` uses the following interface:

```
void shbt_remap(double *col_a, double *col_b, double c, double s, size_t n);
```

Each 512-bit vector processes eight `double` lanes; the scalar tail handles the remaining $n \bmod 8$ elements. For $c^2 + s^2 = 1$, the transformation is

$$a'_i = c a_i + s b_i, \quad b'_i = c b_i - s a_i.$$

The transformation preserves the combined two-column norm in exact arithmetic. The implementation uses binary64 arithmetic, with roundoff included in numerical verification. Listing 1 includes both the vector loop and scalar tail.

8.6 Lock-free communication and finite storage

An SPSC ring with one empty slot publishes a written payload with a release store of its producer index; the consumer uses an acquire load before reading it. Producer and consumer own different indices, and their updates must be atomic. An MPSC ring must distinguish reservation from publication, typically with an individual sequence counter for each slot; one CAS on a shared index

alone does not prevent an uninitialized payload from becoming visible. Ring capacity, counter-wrap convention, lock-free atomic support, full/empty policy, interrupt reentrancy, and progress guarantees belong in the hardware/software contract. An MPSC qualification run shall exercise concurrent publication, including a workload of four producer threads and 4000 messages, and check ordering and loss explicitly. No allocator avoids one source of variability but cannot prove V-33's 5 ps dispatch jitter.

9 Recovery timing, telemetry, and independent safety interlocks

9.1 V-27 maximum recovery-time standard

V-27 defines the primary recovery-time design and verification standard. For a recoverable quench, the complete event-to-healthy-output latency shall satisfy

$$t_{\text{rec}} \leq 120.00 \text{ ns.} \quad (27)$$

The four-stage CPU allocation is specified below. Each stage shall include the software instructions, device transactions, and physical response assigned to it.

Complete recovery stage	Maximum cycles	Maximum ns at 2.5 GHz
Fault inspection and quarantine	40	16.00
Drive blanking and completion	65	26.00
FIFO flush and ECC service	115	46.00
PLL reacquisition and healthy enable	80	32.00
Total	300	120.00

For sequential complete stages with $f_{\text{CPU}} \geq 2.5 \text{ GHz}$,

$$t_{\text{rec}} \leq \frac{40 + 65 + 115 + 80}{2.5 \text{ GHz}} = 120.00 \text{ ns.}$$

The allocation shall include interrupt entry/exit, channel selection, FIFO flush, ECC service, completion reads, polling, and sensor/actuator delays. Platform qualification shall establish the cycle-to-time mapping at every supported clock, including physical delays that do not scale with CPU frequency. The elapsed-time counter shall use its calibrated frequency, which need not equal the CPU frequency. Faults that cannot recover shall terminate in a blanked, latched state; a safe timeout is not a successful V-27 recovery trial.

Theorem 9.1 (Conditional V-27 composition bound). *If four sequential complete stages satisfy cycle bounds (40, 65, 115, 80) at a clock rate of at least 2.5 GHz, with all physical and software delays included, their total elapsed time is at most 120.00 ns.*

Proof. The sum of the stage bounds is 300 cycles. At 2.5 GHz one cycle lasts 0.4 ns, so the total is at most $300(0.4) = 120.00 \text{ ns}$. $\square$

This theorem establishes the composition of the specified bounds. Target timing verification shall separately establish those bounds for the deployed binary, MMIO bus, and physical device.

Listing 2: Z3 conditional composition certificate with executable assertions.

```
from z3 import Ints, Real, RealVal, Solver, Sum, sat, unsat

stages = Ints("inspect blank ecc reacquire")
clock = Real("clock_ghz")
contract = Solver()
for stage, maximum in zip(stages, [40, 65, 115, 80]):
    contract.add(stage >= 0, stage <= maximum)
contract.add(clock >= RealVal("2.5"))
```

```

assert contract.check() == sat
contract.push()
contract.add(Sum(stages) > RealVal("120.00") * clock)
assert contract.check() == unsat
contract.pop()
print("V-27 composition verified under the complete-stage assumptions")

```

The script checks that the assumptions are satisfiable before testing a violation of the V-27 maximum. Execution requires Z3; only an `unsat` violation query establishes this conditional model result.

9.2 Eigenvector rigidity: observable, precision, and latency

The proposed diagnostic is

$$\delta_\Phi = 1 - |\langle \Phi(t), \Phi(t - \tau) \rangle|^2, \quad \|\Phi\| = 1, \quad \delta_\Phi < 10^{-12}, \quad \tau = 1.14 \text{ ns}.$$

The dimensions of Φ must match its Gram matrix: $W^\dagger W$ acts on the bulk code domain, not automatically a 312-dimensional boundary space. More importantly, an exact isometry has $W^\dagger W = I$, which is spectrally degenerate. Its leading eigenvector is not unique and its eigenvector gap is not $1 - \varepsilon_W$. A perturbative bound $\sin \angle(\Phi, \Phi') \lesssim \|\Delta G\|/\text{gap}$ needs an isolated eigenvalue and a measured gap. Monitoring a calibrated invariant subspace or projector is preferable when the spectrum is degenerate; a changing basis inside that subspace is not necessarily a fault.

Binary32 arithmetic has unit roundoff $2^{-24} \simeq 5.96 \times 10^{-8}$. Subtracting a near-unit binary32 overlap from one cannot robustly resolve 10^{-12} . Full double-precision storage and accumulation, normalization checks, compensated reduction, and uncertainty bounds are required even for a proposed implementation of that threshold. A computed value must satisfy $\hat{\delta}_\Phi + u_\delta < 10^{-12}$ to certify health; an ambiguous interval should fail safe, not be silently rounded to zero. This does not guarantee that physical measurement noise is that small.

The ten-load/two-stream AVX-512 reduction requires at least five issue cycles for ten FMAs at two FMAs/cycle, before loads, dependencies, and horizontal reduction. Qualification shall include these costs in the monitor timing budget. The 1.14 ns interval corresponds to 3.42 cycles at 3 GHz and shall remain a design target pending an implementation with demonstrated timing. A dedicated parallel monitor shall provide its RTL, precision, sample-rate, transport, and end-to-end timing analysis. Monitor latency and per-word ECC latency shall be verified independently.

9.3 The 2.50 ns emergency electro-optic shunt

The nominal component budget is

$$t_{\text{shunt}} = 0.33 + 1.25 + 0.42 + 0.50 = 2.50 \text{ ns},$$

for comparator/decision, posted MMIO, interposer propagation, and electro-optic response. A 62 mm trace with $\varepsilon_{\text{eff}} = 2.63$ instead gives approximately 0.335 ns at $c/\sqrt{\varepsilon_{\text{eff}}}$; any 0.42 ns allocation should explicitly include connector and package delay. A posted write completing at the CPU is not proof that an optical field has been attenuated. Propagation, rise time, attenuation ratio, and detected quench-to-safe-output latency must be measured at defined physical endpoints.

The safety circuit shall combine independent overtemperature detection, a latched fault OR, and a normally blanked electro-optic bias clamp. Rigidity-monitor failure, stale samples, clock failure, uncorrectable ECC, and watchdog timeout shall assert blanking; reset shall require healthy physical inputs and an authorized acknowledgement. Candidate overtemperature thresholds are 4.5 K and 8.0 K. Until a hazard analysis selects a validated threshold, the interface treats the comparator output as an independent unsafe input and does not silently adopt the higher value. Hardware shall not allow software to clear it.

For a fault that deposits power P_f , a shunt only bounds extra incident energy by $\Delta Q \leq \int_0^{t_{\text{safe}}} P_f(t) dt$ under a specified power bound. It does not constrain steady-state refrigeration load

or all single-point faults by itself. The 1.14 ns sampling interval, measurement latency, worst-case sample phase, and shunt latency must be added where sequential; the 2.50 ns actuation target is not an unconditional quench-to-safe bound.

10 Complete verification and validation matrix

The following six-column matrix is the primary design and verification standard for all 35 requirements V-01–V-35. Each implementation shall meet its applicable acceptance criteria and verification protocol. V-09 shall be evaluated as an EOM modulation frequency; V-06–V-08 shall use the stated angular-to-ordinary-frequency conversions. After mitigation, the affected requirements and their coupled thermal, timing, and fidelity budgets shall be retested against the same limits. All rows currently have hardware status **qualification pending**; software checks and conditional formal results are summarized separately below.

Table 2: Complete V-01–V-35 requirements, protocols, and mitigation paths.

ID	Description	Metrology instrumentation	Protocol / test procedure	Pass/fail criterion	Mitigation fallback path
V-01	Canonical WZW levels and charge ledger	Exact affine calculation; calibrated sector-resolved correlations	Verify (26, 8, 312) locks and specify the physical observable and uncertainty	$\Delta_{\text{fr}} = 0$; $c_{\text{vis}} = 1325/154$, $c_{\text{parent}}^{\text{comp}} = 351/8$, $c_{\text{dark}} = 1197103/362670$	Revise sector encoding/calibration; a spectrum alone is not a CFT certificate
V-02	Higher-spin truncation error	Multimode quantum tomography	Spectral power sum for $s \geq 5$	$\sum \ W\ ^2 \leq 1.5 \times 10^{-8}$	Increase Voronoi cell density
V-03	Completed-partition modular residual	Sector character calculation plus calibrated observable reconstruction	δ_{mod} versus loss and thermal occupation; no scalar chiral-weight substitution	$\delta_{\text{mod}} \leq 1.2 \times 10^{-7}$ for the completed partition function	Verify modular completion and transfer calibration
V-04	Spatial noise correlation length	Multichannel noise cross-correlation	Spatial correlation fit versus ξ	$\xi_{\text{noise}} \leq 15.0 \mu\text{m}$	Deep-trench phononic barriers
V-05	Boundary isometry operator norm	Operator state tomography	Reconstruct $\ W^\dagger W - P_{\text{code}}\ _{\text{op}}$ over 312 channels, including uncertainty; separately check the fit using <code>formal/formal_verification.py</code>	$\varepsilon_W \leq 3.430 \times 10^{-3}$	Nonunitary channel recalibration or explicit recovery projection; unitary remap alone cannot improve defect
V-06	Intra-hexamer coupling $J_{\text{ring}}/(2\pi)$	High-resolution optical vector analyzer	S_{21} transmission resonance splitting	$45.20 \pm 0.50 \text{ MHz}$	Adjust thermo-optic micro-trim
V-07	Inter-hexamer parasitic crosstalk	Cross-transmission isolation metrology	Adjacent-hexamer injection ratio	$J_{\text{cross}}/(2\pi) \leq 1.25 \text{ MHz}$	Increase spatial isolation pitch
V-08	Kerr anharmonicity χ	Power-dependent resonance shift	Single-photon $\chi/(2\pi)$ shift	$-220.0 \pm 2.5 \text{ MHz}$	Adjust multiple-quantum-well confinement volume
V-09	Synthetic EOM modulation frequency	Optical spectrum analyzer	Sideband amplitude distribution	$14.50 \pm 0.01 \text{ GHz}$	Re-tune RF EOM drive amplitude
V-10	DRAG gate duration and shape	Real-time 40 GSa/s oscilloscope	Pulse-envelope rise/fall fitting	$t_g = 14.21 \pm 0.05 \text{ ns}$	Calibrate FIR predistortion
V-11	Subspace leakage rate	Randomized benchmarking RB-10 ⁶	Continuous Clifford sequence	$E_{\text{leak}} \leq 5.0 \times 10^{-5}$	Optimize DRAG derivative gain
V-12	QEC fault-tolerance threshold	Monte Carlo syndrome injection	Error-injection versus recovery sweep	$\varepsilon_{\text{th}} \geq 1.80 \times 10^{-2}$	Increase decoder bond dimension χ
V-13	Operational logical error rate	10 ⁹ real-time syndrome cycles	Surface-code defect tracking rate	$E_{\text{log}} \leq 2.0 \times 10^{-5}$	Increase syndrome averaging N

ID	Description	Metrology instrumentation	Protocol / test procedure	Pass/fail criterion	Mitigation fallback path
V-14	MPS decoder frame latency	Hardware logic-state analyzer	Raw syndrome to correction vector	$t_{\text{dec}} \leq 150.0 \mu\text{s}$	Scale VPU clock to 1.0 GHz
V-15	Decoder cryo-SRAM bandwidth	BIST memory-throughput counter	Simultaneous read/write streaming	$\text{BW} \geq 14.0 \text{ TB/s}$	Activate interleaved SRAM bank
V-16	2EE2P adiabatic bit dissipation	Cryogenic calibrated power meter	Transition energy per line at 4.2 K	$E_{\text{bit}} \leq 9.0 \times 10^{-18} \text{ J}$	Lengthen adiabatic ramp τ_r
V-17	Total control dynamic power	Multi-channel DC power analyzer	11776 lines active at 877 MHz	$P_{\text{dyn}} \leq 95.0 \text{ mW}$	Enable subcluster clock gating
V-18	Aerogel core refractive index	Spectroscopic ellipsometry at 4.2 K	Refractive index at 1550 nm	1.0182 ± 0.0008	Optimize sol-gel aging time
V-19	Waveguide sidewall angle	Cross-sectional FE-SEM metrology	Etch-profile angle	$89.88 \pm 0.05^\circ$	Adjust ICP Cl_2/CH_4 gas ratio
V-20	Interfacial delamination toughness	Cryogenic four-point bend delamination test	Mode-I strain energy G_{Ic}	$G_{Ic} \geq 3.50 \text{ J/m}^2$	Introduce ALD Al_2O_3 seed layer
V-21	Shielding residual magnetic field	Cryogenic quantum SQUID metrology	Maximum residual field in active volume	$ B \leq 0.0975 \text{ nT}$	Degauss Cryoperm-10 shells
V-22	Waveguide penetration RF shielding	40 GHz network-analyzer S_{21} test	Attenuation through 364 penetrations	Attenuation $\geq 140.0 \text{ dB}$	Lengthen cutoff-sleeve ratio L/D
V-23	Interposer characteristic impedance	TDR reflectometry at 4.2 K	40 GHz trace impedance	$50.12 \pm 0.80 \Omega$	Adjust copper-etch compensation
V-24	Trace far-end crosstalk (FEXT)	40 GHz multiport VNA metrology	S_{41} intertrace coupling ratio	$\text{FEXT} \leq -70.0 \text{ dB}$	Add ground picket vias
V-25	Total 4.2 K cryogenic heat load	Precision helium-boiloff calorimeter	Validate 25% front-end duty, idle/wake overhead, absorbed DCE pump inside driver allocation, and converter heat	$P_{\text{tot}} \leq 850.0 \text{ mW}$; 819.44 mW baseline plus at most 30.56 mW unallocated	Reduce duty, absorption, or restage electronics with measured conduction; no double counting
V-26	Thermal headroom capacity margin	Cryostat calibrated-heater loading	Margin against 1.500 W limit	Margin $\geq 40.0\%$	Engage secondary pulse-tube head
V-27	Post-quench recovery time	Nanosecond time-interval counter and binary WCET analysis	Four-step recovery in <code>kernel/src/shbt_core_runtime.c</code> ; stage maxima (40, 65, 115, 80) cycles; functional tests plus the formal composition check in Listing 2	$t_{\text{rec}} \leq 120.00 \text{ ns}$ at $f_{\text{CPU}} \geq 2.5 \text{ GHz}$; no omitted delays	Reduce and measure complete-stage maxima; unrecoverable faults fail closed, not pass
V-28	Core RMS vibration displacement	Laser Doppler vibrometer at 4.2 K	Integrated 1–500 Hz displacement	$x_{\text{rms}} \leq 0.85 \text{ nm}$	Increase piezo feed-forward gain

ID	Description	Metrology instrumentation	Protocol / test procedure	Pass/fail criterion	Mitigation fallback path
V-29	28 nm FD-SOI cryo threshold shift	Cryo-prober parameter analyzer	V_{th} at forward well bias 1.2 V	$\Delta V_{th} \leq 200$ mV	Trim back-gate charge pump
V-30	Baseband DAC effective resolution	Audio Precision / high-speed FFT	ENOB at 1.2 GSa/s and 4.2 K	ENOB ≥ 14.8 bits	Calibrate INL/DNL lookup table
V-31	Heterodyne readout fidelity	Single-shot state classification	Integrated 12.0 ns readout window	$F_{read} \geq 99.95\%$	Increase LO optical power
V-32	Master-clock integrated jitter	Agilent E5052B signal-source analyzer	Phase-noise integral, 10 Hz–10 MHz	$\sigma_t \leq 8.50$ fs	Clean DRO power-supply rail
V-33	Bare-metal runtime latency jitter	Hardware high-speed scope trigger	Command-dispatch jitter	$\delta t \leq 5.0$ ps	Lock SRAM vector pipeline
V-34	SECDED Hamming(72,64) ECC decode time	C microkernel <code>kernel/src/shbt_core_runtime.c</code> / VPU cycle-accurate simulator / ELA	Correction-cycle duration; single-bit correction and double-bit detection tested using <code>tests/reference_test.c</code> and <code>tests/run_all_tests.py</code>	$t_{ecc} \leq 1.20$ ns	Synthesize dual-rail logic
V-35	Wafer net channel module yield	Automated optical/electrical probe	Functional channels at least 312 of 336	Module yield $\geq 85.00\%$	Enable backup-channel remapping

10.1 Implemented verification summary

The implementation provides the following requirement-specific verification interfaces:

- V-05: `formal/formal_verification.py` checks the 312-channel isometry-fit value 3.42640×10^{-3} against the ceiling 3.430×10^{-3} ; physical encoding requires independent tomography.
- V-27: the four-step recovery interface uses the complete-stage cycle envelope (40, 65, 115, 80). Listing 2 specifies the formal composition check for $t_{\text{rec}} \leq 120.00 \text{ ns}$ at $f_{\text{CPU}} \geq 2.5 \text{ GHz}$.
- V-34: `kernel/src/shbt_core_runtime.c shbt_ecc_decode` implements SECDED Hamming(72,64) correction/detection; the 1.20 ns budget requires target-VPU timing verification in addition to functional tests.

The master runner `tests/run_all_tests.py` sequences the C reference test, multi-physics FEA, photonic CAD, HIL/MPS-QEC, pytest, Z3 formal verification, and EDA integrity checks, returning exit code 0 when all suites pass.

10.2 Evidence-to-requirement discharge rules

For V-01–V-05, acceptance shall use requirement-specific estimators of the boundary algebra, modular residual, noise correlation, and encoding defect, with statistical and systematic uncertainties. For V-06–V-11, the calibrated Hamiltonian and pulse-transfer model shall match the physical configuration. For V-12–V-15, decoder correctness, logical error, memory throughput, and latency shall be measured separately. For V-16–V-20 and V-23–V-29, verification shall include consistent arithmetic, coupled physical budgets, and calibrated measurements. V-21, V-22, V-30, and V-32 shall include dedicated shielding, converter-linearity, and clock-noise characterization. V-31 shall include readout-model validation and measured classification fidelity; V-33/V-34 shall include target-hardware timing; V-35 shall include complete-module statistics. A requirement shall pass only when its defined protocol and acceptance criterion are satisfied; model predictions shall be recorded separately from physical qualification evidence.

11 Reliability, cold boot, and bill of materials

11.1 FMEDA model and reliability allocation

Subsystem	Base FIT	Safe/detected FIT	Dangerous undetected FIT	Nominal DC (%)
Photonic array	45.2	44.1	1.1	97.57
FD-SOI drivers	112.5	110.8	1.7	98.49
MPS decoder	88.4	87.2	1.2	98.64
DRO/PLL clock	24.1	23.8	0.3	98.76
Pulse-tube cooler	120.0	116.5	3.5	97.08
Shielding	22.3	22.1	0.2	99.10
Aggregate model	412.5	404.5	8.0	98.06

One FIT means 10^{-9} h^{-1} . The base-rate sum is 412.5 FIT, corresponding to $10^9/412.5 = 2.42424 \times 10^6 \text{ h}$ under a constant-rate model. The row sums of the next two columns are 404.5 and 8.0, giving an aggregate detected fraction $404.5/412.5 \simeq 98.06\%$. Qualification shall evaluate this model result separately from the overall diagnostic-coverage objective. Additional design objectives are SFF 96.81%, overall diagnostic coverage 98.41%, SIL-3 qualification, and a cosmic-neutron upset rate $9.1 \times 10^{-9} \text{ h}^{-1}$ per system. An aggregate detection ratio or an MTBF does not establish a safety-integrity qualification. Qualification requires safe/dangerous failure classification, demand mode, proof-test coverage, a common-cause model, and the applicable certification process; the design objectives alone do not establish compliance.

11.2 Six-phase cold boot

Phase	State/exit condition	Nominal duration
1	Vacuum evacuation to $P < 10^{-6}$ mbar	4 h
2	Cooldown 300 K to 4.2 K, $ \dot{T} < 1.5$ K/min	6 h
3	PDN and FD-SOI back-gate calibration, +1.2 V	2 min
4	Photonic interconnect / 40 GHz DRO phase lock	30 s
5	Microcavity tuning and four-bit DAC offset nulling	45 s
6	Boundary-isometry characterization and QEC activation	15 s

Every phase shall define an exit predicate, timeout, calibration identifier, and safe rollback. Phase completion shall depend on the exit predicate rather than elapsed nominal duration. Optical enabling shall require passing vacuum, temperature, lock, ECC, channel-yield, and control self-tests. The boot record shall contain measured channel-lock status, cold-stage power, timestamps, and the result of each exit predicate.

11.3 Bill of materials

Table 3: Ten-item BOM with candidate suppliers and design specifications.

ID	Subsystem	Part / drawing	Supplier	Qty	Design specification
01	Photonic hexamer wafer	SHBT-WAFER-INP-REV5	Custom / AIXTRON MOCVD fab	1	High- Q InP/InGaAsP at 4.2 K
02	12-layer RF interposer	SHBT-INT-R4350B-12L	Rogers / custom fab	1	Ultra-low loss at 4.2 K
03	28 nm FD-SOI UTBB ASIC	SHBT-ASIC-FDSOI-28-CRYO	GlobalFoundries / custom	4	Forward back-biased at 4.2 K
04	Cryogenic MPS decoder	SHBT-DEC-MPS-128VPU	Custom cryo-CMOS foundry	2	14.3 TB/s SRAM at 4.2 K
05	Cryoperm-10 shield shells	SHBT-MAG-CP10-NESTED	Vacuumschmelze GmbH	2	$\mu_r = 145000$ at 4.2 K
06	Bulk YBCO cylinder	SHBT-MAG-YBCO-MEISSNER	Can Superconductors	1	$J_{c0} = 2.4$ MA/cm ²
07	Two-stage pulse-tube cooler	PT420-RM-1.5W	Cryomech / Bluefors	1	1.500 W at 4.2 K baseline
08	40 GHz cryogenic DRO	DRO-40G-CS-CRYO	Synergy Microwave / custom	1	6.42 fs jitter
09	Stainless semirigid coax	UT-047-SS-SS-CRYO	Carlisle Interconnect	104	0.047-inch, 4.2 K
10	Hermetic SMF-28 fiber array	FA-364-SMF28-HERM	Diamond SA / custom	1	Hermetic feedthrough array

Supplier names and drawing numbers identify candidate components, not confirmed availability or qualified products. The separate DCE substrate, sapphire boule, acoustic matching layer, and conversion interface need controlled BOM entries and acceptance drawings before the consolidated architecture is procurement-ready.

12 Conclusion and release gates

The SHBT-R specification integrates the canonical (26,8,312) boundary algebra, 312-channel photonic processor, squeezed-state seeding, cryogenic interfaces, digital twin, and `shbt-os` runtime

into one architecture. Its equations, reference implementations, interface contracts, and V-01–V-35 matrix define the implementation and acceptance baseline. The central outstanding gates are a physical encoding of the affine sectors and seed-to-optical conversion chain; a consistent isometry calibration and recovery channel; a validated entropy/power budget; a specified braid representation and high-precision compiler certificate; and a versioned hardware interface with measured end-to-end timing. Release requires completion of these gates and requirement-level verification; software and model-level results do not substitute for physical quantum-computer qualification.

References

- [1] SHBT Precision Boundary Simulator, software repository, <https://github.com/syslown/shbt-precision.git>.
- [2] SHBT-R Microkernel & Digital-Twin Engine, software repository, <https://github.com/syslown/shbt-qc.git>.
- [3] A. Sekine, R. Murakami, and Y. Doi, “Microwave-to-Optical Quantum Transduction of Photons for Quantum Interconnects”, arXiv:2509.26349 (2025).
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- [9] Intel Corporation, “Intel AVX-512 Instructions”, technical article, 20 June 2017; and “Floating-Point Reference Sheet for Intel Architecture”.
- [10] N. Bjørner, L. de Moura, L. Nachmanson, and C. Wintersteiger, “Programming Z3”, Z3 project technical tutorial.
- [11] R. Simon, “Peres–Horodecki separability criterion for continuous variable systems”, *Physical Review Letters* **84**, 2726 (2000), arXiv:quant-ph/9909044.

A Code Availability

A.1 SHBT Precision Boundary Simulator

The SHBT Precision Boundary Simulator provides the Rust/Python boundary-sector implementation [1]:

<https://github.com/syslown/shbt-precision.git>

```
git clone https://github.com/syslown/shbt-precision.git
```

The repository contains the Rust library in `src/`, its `Cargo.toml` manifest, Python simulation entry points, and tests. Follow its build instructions for the Rust library and Python bindings.

A.2 SHBT-R Microkernel & Digital-Twin Engine

The SHBT-R Microkernel & Digital-Twin Engine provides the engineering simulator, bare-metal `shbt-os` runtime, HIL integration, formal checks, and EDA export tools [2]:

<https://github.com/syslown/shbt-qc.git>

```
git clone https://github.com/syslown/shbt-qc.git
cd shbt-qc
python3 cli/main.py build-kernel
python3 cli/main.py verify
python3 tests/run_all_tests.py
```

Run these commands with the repository’s documented dependencies installed and an AVX-512-capable execution target for the corresponding C tests. The build command produces the hosted shared object `shbt_reference.so`; this artifact supports simulation and does not replace a target-linked bare-metal image.

A.3 Implementation and verification interfaces

Within `shbt-qc`, `kernel/src/shbt_core_runtime.c` implements the C runtime, `kernel/src/shbt_user_mmio.c` provides the hosted MMIO shim, and `kernel/include/shbt_hardware.h` defines the register interface. The linker script `kernel/linker.ld` allocates the 2112-byte arena on a 64-byte boundary. The hosted test driver `tests/reference_test.c` exercises SECDDED correction/detection, recovery, and remapping. Formal contracts are implemented in `formal/formal_verification.py`; the master runner is `tests/run_all_tests.py`. Record the repository commit, configuration, toolchain, execution target, and test output for each verification run. Associate results with the applicable V-01–V-35 requirements and their assumptions; functional tests and formal bounds do not establish physical-device timing or fabrication qualification.

A.4 Self-contained manuscript build

Build `qc.pdf` from the directory containing `main.tex`:

```
pdflatex -interaction=nonstopmode -halt-on-error -jobname=qc main.tex
pdflatex -interaction=nonstopmode -halt-on-error -jobname=qc main.tex
```

Repeat the command if the log requests another pass, until citations, cross-references, the table of contents, and table widths are stable. The `-jobname=qc` option sets the output basename; a trailing `-o qc.pdf` argument does not select the PDF filename. Alternatively, automate the required passes with:

```
latexmk -pdf -jobname=qc main.tex
```

The manuscript includes its equations, tables, code listings, and bibliography. Compilation requires no external figure files, software checkout, network access, or shell-escape step.

B Supplementary Microcavity and Cluster Specifications

This appendix defines normative supplementary specifications for the microcavity array, cluster organization, and physical interfaces of SHBT-R. Device fabrication, calibration, reconstruction, and digital-twin configuration shall use these channel definitions and the primary design and verification standards in Section 1.4. Table 2 defines the governing acceptance limits; nominal parameters in this appendix shall be evaluated against those limits.

B.1 Channel inventory and cluster addressing

The processor shall operate 312 active InP/InGaAsP microcavity channels in 52 hexamers. The fabrication layout shall provide 336 channels, including 24 spares. For hexamer index $m \in \{1, \dots, 52\}$ and cavity index $j \in \{1, \dots, 6\}$, the active logical channel identifier is

$$q = 6(m - 1) + (j - 1), \quad q \in \{0, \dots, 311\}.$$

Ring neighbors use cyclic indexing $j + 6 \equiv j$. A calibration map shall associate each logical channel with one physical cavity and its resonance, drive, readout, and ECC metadata. Spare activation shall replace a physical assignment while preserving 312 active logical channels and updating all associated calibration maps.

Table 4: Microcavity and cluster design parameters.

Quantity	Specification	Convention
Active channels	$52 \times 6 = 312$	Logical identifiers 0–311
Fabricated channels	336, including 24 spares	At least 312 functional channels per accepted module
Substrate	100 mm semi-insulating InP(100)	InP/InGaAsP microcavity array
Sampling pitch	$a = 12.51 \mu\text{m}$	$k_c = \pi/a = 2.5113 \times 10^5 \text{ m}^{-1}$
Optical resonance	$\omega_0/(2\pi) = 193.41 \text{ THz}$	Approximately 1550 nm
Intracluster hopping	$J_{\text{ring}}/(2\pi) = 45.20 \text{ MHz}$	Six-site cyclic ring
Intercluster hopping	$J_{\text{cross}}/(2\pi) = 1.15 \text{ MHz}$	Calibrated cluster adjacency
Kerr coefficient	$\chi/(2\pi) = -220.00 \text{ MHz}$	Finite computational truncation
Synthetic-mode hopping	$\kappa_{\text{syn}}/(2\pi) = 12.80 \text{ MHz}$	Adjacent frequency bins
EOM modulation	$\Omega_{\text{EOM}}/(2\pi) = 14.50 \text{ GHz}$	Ordinary-frequency spacing
DCE pump and seed	72 GHz drive; two 36 GHz modes	$\Omega = \omega_a + \omega_b$
Operating bath	4.2 K	Sapphire reservoir and aerogel interface

B.2 Hexamer spectrum and calibrated control

The Hamiltonian in Section 3 defines the ring, synthetic-frequency, intercluster, and drive terms. For an isolated uniform hexamer, the six normal modes have frequencies

$$\omega_\ell = \omega_0 - 2J_{\text{ring}} \cos(2\pi\ell/6), \quad \ell = 0, \dots, 5.$$

The ordinary-frequency bandwidth is $4J_{\text{ring}}/(2\pi) = 180.80 \text{ MHz}$. All simulation and calibration inputs used for V-06–V-11 shall use the couplings in Table 4, with angular-to-ordinary-frequency conversions applied explicitly. Drive calibration shall include pulse area, phase, DRAG derivative gain, transfer function, and computational-subspace leakage.

B.3 Boundary sectors and encoding acceptance

The WZW affine levels are $(k_l, k_q, K) = (26, 8, 312)$, with framing locks $K/(2k_l) = 6$ and $K/(3k_q) = 13$. The sector ledger is

$$c_{\text{vis}} = \frac{1325}{154}, \quad c_{\text{parent}} = \frac{351}{8}, \quad c_{\text{dark}}^{\text{comp}} = \frac{1197103}{362670}.$$

These levels specify boundary-sector algebra; the physical channel inventory is specified independently. The encoding map shall satisfy

$$W = WP_{\text{code}}, \quad \left\| W^\dagger W - P_{\text{code}} \right\|_{\text{op}} \leq 3.430 \times 10^{-3}.$$

V-05 acceptance shall require the reconstructed operator-norm defect plus its stated uncertainty to satisfy Equation 24. The channel-count fit in Equation 25 evaluates to 3.4263993×10^{-3} at 312 channels and shall be recorded as a model prediction separately from tomographic acceptance evidence. Unitary remapping preserves the Gram-defect norm; recalibration and recovery projections shall be assessed using the same V-05 limit.

B.4 Squeezed-state seeding and physical interfaces

The seeding stage shall couple a 72 GHz parametric drive to two 36 GHz modes. For squeezing parameter r , the vacuum-input pair population is $\sinh^2 r$; the thermal-input covariance and conversion noise shall be included when evaluating the delivered state. The microwave-optical interface in Section 3 specifies conversion efficiency, bandwidth, added noise, and cryogenic dissipation. Optical seeding shall be evaluated at the microcavity input, including transmission loss and the measured thermal occupancy.

The incident duty-averaged DCE pump allocation is 74.464 mW within the 412.00 mW driver allocation. The total cold-stage baseline is 819.44 mW, leaving 30.56 mW under V-25's 850.0 mW ceiling. All conversion, idle, wake-up, and interconnect loads shall be included without double counting. V-26 requires at least 40% capacity margin relative to the specified 1.500 W cooling capacity.

B.5 Fabrication and module acceptance

The process sequence is MOCVD epitaxy, electron-beam lithography, low-damage ICP-RIE, sol-gel coating, and supercritical drying. The process settings, dimensional targets, and statistical process-control quantities are specified in the fabrication section. Each module shall retain its process-lot identifier, dimensional metrology, optical calibration, spare-channel assignments, and cryogenic acceptance results. V-18–V-20 govern aerogel index, sidewall angle, and interfacial toughness; V-35 requires a module yield of at least 85.00% under the at-least-312-of-336 functional-channel criterion. The remaining electrical, thermal, shielding, readout, and clock requirements are defined by the single V-01–V-35 matrix in Table 2.

C Supplementary Digital-Twin Architecture and Register Maps

This appendix defines normative supplementary specifications for the digital-twin modules, data exchange, SHBT-MMIO-1 register bank, and recovery/ECC integration. All modules shall implement the primary design and verification standards in Section 1.4. The implementation repositories and build entry points are given in Appendix A.

C.1 Module organization and data exchange

Table 5: Digital-twin implementation interfaces.

Module	Repository path	Required outputs
Multiphysics	<code>simulator/multi_physics/thermal_fea_sp_sim.py</code>	Temperature fields, contact flux, displacement/phase spectra, SI/PI response
Photonic CAD and yield	<code>simulator/cad_yield/spc_photonic_cad.py</code>	Calibrated coupling model, tolerance samples, accepted-channel masks
HIL and MPS-QEC	<code>simulator/hil_qec/hil_mps_qec.py</code>	Normalized state, stabilizer outcomes, recovery requests, MMIO traces
Microkernel	<code>kernel/src/shbt_core_runtime.c</code>	ECC results, bounded recovery status, channel remapping
Register interface	<code>kernel/include/shbt_hardware.h</code>	ABI version, aligned register layout, masks and access rules
Formal verification	<code>formal/formal_verification.py</code>	Satisfiable assumptions and invariant-query results
EDA export	<code>eda/exporters/pdk_hexamer_exporter.py</code> ; <code>eda/exporters/export_interposer_rf.py</code>	GDSII and Touchstone S2P artifacts
Build and test	<code>cli/main.py</code> ; <code>tests/run_all_tests.py</code>	Build products and requirement-linked test results

Each run shall record the configuration, calibration identifiers, software commit, random seeds, units, sampling times, and numerical tolerances. Channel identifiers shall follow Appendix B. Thermal and vibration outputs shall feed the photonic/noise model through calibrated transfer functions; decoder and microkernel results shall return through the same timestamped HIL interface.

C.2 Multiphysics and quantum-state contracts

The thermal model shall solve

$$\rho c_p(T) \partial_t T = \nabla \cdot [\kappa(T) \nabla T] + \dot{q}$$

with conservative face fluxes and explicit material and bath boundary conditions. For an interface parameterized by $G_K(T) = \alpha_K T^3$, the finite-temperature contact flux is

$$q_K = \frac{\alpha_K}{4} (T_{\text{hot}}^4 - T_{\text{cold}}^4).$$

Each contact flux shall replace the corresponding bulk face flux and transfer equal and opposite energy between neighboring cells. Mesh/time refinement, energy conservation, positive temperatures, and calibrated contact conductance are numerical acceptance requirements.

The vibration model shall propagate displacement through the optical phase transfer function and integrate the specified 1–500 Hz band. The interposer model shall export impedance, insertion loss, crosstalk, and PDN response with frequency, reference-plane, and termination conventions. Photonic Monte Carlo samples shall use the calibrated microcavity parameters and acceptance windows of the physical design; module yield shall count at least 312 functional channels out of 336.

MPS evolution shall preserve compatible bond dimensions and track normalization and truncation error. Stabilizer measurements shall use contracted expectation values or the specified measurement sampling model. Pauli injection, syndrome extraction, recovery selection, and logical-error estimation shall be tested against exact small-system calculations. Quantum stabilizer syndromes and classical SECDED Hamming(72,64) ECC syndromes shall use separate data types and interfaces. The MPS frame time shall satisfy $t_{\text{dec}} \leq 150.0 \mu\text{s}$ under V-14; ECC decoding shall satisfy $t_{\text{ecc}} \leq 1.20 \text{ ns}$ under V-34.

C.3 SHBT-MMIO-1 address map

SHBT-MMIO-1 shall use base address 0x70000000. The bank shall contain fourteen 32-bit words aligned on four-byte boundaries and accessed as uncached device memory. Its 56-byte span shall occupy 0x70000000–0x70000037, with register-start offsets 0x00–0x34. Table 6 specifies the same normative offsets and access modes as Section 8.2; hardware, firmware, and the virtual bus shall implement this single register contract.

Table 6: SHBT-MMIO-1 register offsets and semantics.

Offset	Register	Access	Required behavior
0x00	status	Read-only	Bit 0 OVERTEMP; bit 1 PLL_LOCK; bit 2 ECC_ERR; bit 3 FAULT_ST. OVERTEMP is active-high unsafe.
0x04	blank	Read/write	Value 1 blanks all active drives; value 0 requests enable subject to healthy physical inputs and interlocks.
0x08	fifo_data	Request / status	Bit 0 requests flush and self-clears; bit 1 indicates busy.
0x0C	pll_ctrl	Request	Write 1 requests phase lock; status bit 1 indicates completion.
0x10	ecc_low	Read/write	Low 32 bits of the selected coherent ECC payload.
0x14	ecc_high	Read/write	High 32 bits of the same payload.
0x18	ecc_check	Read/write	Bits 0–6 store Hamming parity; bit 7 stores overall even parity.
0x1C	ecc_commit	Request	Write 1 atomically commits the corrected payload and check byte.
0x20	fault_latch	Write-one-to-set	A software fault sets the latch; only the independently authorized reset clears it.
0x24	channel_select	Read/write	Logical channel 0–311; completed selection latches a coherent ECC snapshot.
0x28	abi_version	Read-only	Platform binding shall require value 1 and reject other versions before issuing control commands.
0x2C	phase_offset	Read/write	32-bit phase-calibration payload with scaling defined by the calibration record.
0x30	ecc_counts	Read-only	Bits 0–7 count corrected errors; bits 8–15 count uncorrectable errors. Both counters saturate.
0x34	control	Request / payload	Start bit 0; soft reset bit 1; recovery bit 2; pump attenuation bits 8–15; reserved bits zero.

Every command shall support a side-effect-free completion read. The virtual bus shall implement the same command completion, self-clearing requests, ECC snapshot, atomic commit, and saturating counters as the hardware. A software reset shall not clear an unsafe physical condition or an independently latched fault.

C.4 ECC transaction and recovery sequence

The runtime shall select a logical channel, complete the selection write, and read the latched 64-bit payload and eight check bits. SECDED Hamming(72,64) shall correct one erroneous data or parity bit and detect every two-bit error under the stated fault model. Corrected data and check bits shall be written before `ecc_commit`; an uncorrectable word shall remain uncommitted and shall initiate fail-closed handling. The syndrome decision table is in Section 8.3.

The complete post-quench sequence shall inspect and quarantine the fault, assert drive blanking, flush and service the ECC transaction, and request phase-lock reacquisition before enabling healthy outputs. The end-to-end requirement is

$$t_{\text{rec}} \leq 120.00 \text{ ns}, \quad (c_1, c_2, c_3, c_4) \leq (40, 65, 115, 80), \quad f_{\text{CPU}} \geq 2.5 \text{ GHz},$$

where the cycle limits apply componentwise to complete stages. V-27 verification shall measure the complete event-to-healthy-output latency. The allocation includes interrupt overhead, bus completion, polling, ECC, and physical response. The cycle-composition proof and its implementation assumptions are given in Section 9. A persistent fault or timeout shall leave drives blanked and the fault latched. Verification shall record this as a fail-closed outcome separately from a successful V-27 recovery.

C.5 Runtime memory and verification outputs

The `shbt-os` arena shall occupy 2112 payload bytes aligned to 64 bytes: 640 bytes for 160 binary32 active-register entries and 1472 bytes for ledger records and command descriptors. Protecting every 64-bit payload word requires 264 additional ECC bytes unless the memory controller stores check bits out of band. Stacks, instruction storage, and external matrices shall have separate linker allocations. The hosted MMIO shim shall expose the same ABI as the bare-metal register binding.

Verification outputs shall identify the applicable V-01–V-35 requirement, configuration, acceptance criterion, observed value, uncertainty, and pass/fail result. Functional tests shall cover all 72 single-bit faults and all $\binom{72}{2} = 2556$ distinct double-bit faults per test codeword, invalid channel selection, read-only-register protection, command completion, persistent faults, and timeout behavior. Formal checks shall first establish satisfiable assumptions and then require `unsat` for the forbidden-behavior query. Hardware release shall additionally provide calibrated physical measurements and target timing.